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Project Document template for projects financed by the various GEF Trust Funds

Project title: San Salvador Low-emission Urban Development Path		
Country: El Salvador	**Implementing Partner (GEF Executing Entity):** **National Energy Council – CNE (implementation)**	**Execution Modality:** **National Implementation Modality (NIM)**
Contributing Outcome (UNDAF/CPD, RPD, GPD): CPD 2016-2020, 2.6 Measures have been taken for reliable, sustainable and efficient energy use. Indicators: (i) Metric tons of greenhouse gas emissions avoided. (ii) Number of entities taking integral low-carbon development measures.		
UNDP Social and Environmental Screening Category: C (High)	**UNDP Gender Marker: 2**	
Atlas Award ID: 00107731	**Atlas Project/Output ID: 00107946**	
UNDP-GEF PIMS ID number: 5462	**GEF Project ID number: 9038**	
LPAC meeting date: 17 January 2020		
Latest possible date to submit to GEF: N/A		
Latest possible CEO endorsement date: N/A		
Planned start date: 1 June 2020	**Planned end date: 31 May 2025**	
Expected date of Mid-Term Review: **25 November 2022**	**Expected date of Terminal evaluation:** **28 February 2025**	
Brief project description: The Project aims to introduce low-emission urban mobility concepts and energy efficiency management strategies among the municipalities composing the San Salvador Metropolitan Area (AMSS), thereby reducing national dependency on imported oil derivatives and combating energy sector GHG emissions. To this purpose, the Project will strengthen the institutional and legal framework for low-emission mobility solutions (including the Bus Rapid Transit SITRAMSS) and implement a trust fund for EE investment in the public sector (FIDEnergetica). It will develop technical standards and quality criteria as input for regulation and the design of demonstration pilots. Capacity building in the field of urban planning is addressed by fostering international partnerships and inter-sectorial working groups. The expected fuel savings (43,666 GJ annually) will benefit approx. 50,000 individuals (m/f: 40%/60%) and translates into fiscal budget savings of US$ 1.27 mln annually. EE measures by municipalities will translate into annual electricity savings (3,776 MWh annually) and budget savings of US$ 640,000. The estimated total GHG emission reductions over a 10-year period are about 67 kton CO2eq (direct) and 195 kton CO2e (post-project).		

(1) FINANCING PLAN	
GEF Trust Fund	USD 2,420,548
(1) Total Budget administered by UNDP	**USD 2,420,548**
(2) CONFIRMED CO-FINANCING	
Ministry of Environment and Natural Resources	USD 500,000
National Energy Council	USD 12,000,000
Ministry of Public Works, Housing and Urban Development	USD 17,671,904
San Salvador Metropolitan Area Office of Planning	USD 3,800,000
Municipality of Santa Tecla	USD 500,000
UNDP	USD 130,000
(3) Total confirmed co-financing	**USD 34,601,904**
(4) Grand-Total Project Financing (1)+(2)	**USD 37,022,452**

SIGNATURES:

NOTE: IF THE PROJECT DOCUMENT IS IN FRENCH OR SPANISH, THE FINAL PROJECT DOCUMENT MUST BE CLEARED BY THE RTA BEFORE SIGNATURE.

Signature: print name below	**Agreed by Government Development Coordination Authority**	**Date/Month/Year:**
Signature: print name below	**Agreed by Implementing Partner**	**Date/Month/Year:**
Signature: print name below	**Agreed by UNDP**	**Date/Month/Year:**

Key GEF Project Cycle Milestones:
Project document signature: within 25 days of GEF CEO endorsement
First disbursement date: within 40 days of GEF CEO endorsement
Inception workshop date: within 60 days of GEF CEO endorsement
Operational closure: within 3 months of posting of TE to UNDP ERC
Financial closure: within 6 months of operational closure

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II. DEVELOPMENT CHALLENGE

Introduction

1. The Republic of El Salvador, with an area of 21,040 km^2, is the smallest country in Central America. It shares borders with Guatemala in the west and Honduras in the north and east. El Salvador has a total population of 6.6 mln people[1] which translates into an average population density of 313 hab/km^2, of which 3.5 mln (53%) are women and 3.1 mln (47%) are men. Life expectancy at birth is 73.5 years (CEPAL, 2017). Migration has traditionally been a mechanism to cope with the scarcity of land and the poor living conditions in the rural areas. Presently an estimated 2.5 mln nationals live in the USA, whose remittances are a major source of income for the country.
2. The territory is politically divided into 14 departments and 262 municipalities. According to the 1983 Constitution, the municipalities are autonomous and governed by a municipal council, while the departments respond to the central Government. The municipalities have technical and budgetary autonomy but must cooperate with the central Government and its agencies. The municipalities are organized into the Council of Salvadoran Municipalities (COMURES) which is a not-for-profit association under private law. Decentralization of the State is seen as a way forward to improve governance and reduce the disparities between the regions in terms of human development and infrastructure, such as through the National Territorial Planning and Development Law (LODT, 2011). The process however is hampered by capacity deficits including fiscal budget constraints and the limited financial means of the municipalities themselves allocated through the FODES.[2]
3. While El Salvador is still predominantly rural (60.2% of population), the Metropolitan Area of San Salvador (AMSS) is the political, financial, economic and cultural center of the country. It concentrates 1.7 mln people (25.7% of the population) and 70% of public and private investment on an area of just 3% of the national territory. After the 1986 earthquake, the municipalities composing the AMSS set up a voluntary commonwealth (*mancomunidad*) to promote a common territorial development agenda, by creating the Council of the AMSS (COAMSS) and its Planning Office (OPAMSS) in 1993.[3] In 2015 the Metropolitan Development Council

[1] Total 6,581,860 people according to the Multi-purpose Household Survey (EHPM, 2017).

[2] FODES = Fund for Economic and Social Development of Salvadoran Municipalities. Through FODES, 8% of fiscal budget is passed to the municipalities.

[3] In 1993, the *Ley de Desarrollo y Ordenamiento Territorial del Área Metropolitana de San Salvador y de los Municipios Aledaños*

(CODEMET) was installed as the coordinating organ between the AMSS and the Central Government; and in 2016, OPAMSS delivered the AMSS Master Plan (*Esquema Director*). Both were already foreseen by Law in 1995 but were not pushed forward due to a lack of political momentum and financial resources.

4. With a GDP of USD 26,797 mln, El Salvador is nowadays classified as a lower middle-income country. Since the early 1990s, a structural transformation of the economy has taken place from agriculture and livestock towards the provision of services. As of 2016, GDP was structured as follows: commerce, hotels and restaurants (20.6%); manufacturing industry (19%); agriculture and livestock (10.1%); and public services (8.9%). Economic performance (2.4% annual growth) falls far behind the average for middle-income countries (5.4%) and is below the average for the region (3.9% Central America and 2.9% for Latin America). This is due to a series of structural problems including low investment in productive systems and infrastructure, and high consumption of imported goods fuelled by the remittances received from abroad.
5. Social programmes have contributed to combat poverty levels from 39.9% (2008) to 29.2% (2017), while extreme poverty was reduced from 12.4% to 6.2%. Significant progress has been made over the last decades including vaccination levels (76%, 1990; 93%, 2016), access to drinking water (79%, 1990; 88.3%, 2016) and improved sanitation (56%, 1990; 97.9%, 2016). Multi-dimensional poverty (MDP) levels remain high however (33.4%, equivalent to 611,480 households, in 2017) and make evident the difference in living conditions between rural (53.3% MDP) and urban areas (21.1%). Factors contributing to MDP include: food insecurity (20.9%), no access to drinking water (21.1%), and sanitation (45.8%).
6. Social insecurity and violence by gangs (the so-called *maras*) is perceived by many as the most urgent development challenge for El Salvador. Homicide rates in El Salvador are among the highest in the world (61 violent deaths per 100,000 people in 2014.[4] One of the main drivers behind gang violence is the lack of opportunities for low-income young people, especially male. The EHPM 2014[5] revealed that 26.6% of individuals between 15-24 years do not study or work; under low-income households, this figure increases to 48.5%. Formal unemployment in El Salvador is only 7% but 65.8% of the working population (nearly 2/3) is underemployed or works intermittently. As a result, only 1/3 (34%) is covered by some form of social security, to be differentiated between urban (43.1%) and rural areas (18.1%). Weak presence of the Government outside the main districts leaves space for the gangs to take control of peripheral areas, such as happens in the AMSS.

Climate change

7. In 2010, El Salvador was categorized as the country most vulnerable to climate change worldwide: 88.7 % of the territory, hosting 95.4% of the population, has been classified as risk area. Between 2009-2011, El Salvador was struck by five extreme climate events[6] which caused economic losses of more than USD$ 1,300 mln in particular affecting the infrastructure supporting productive systems, including the road network. The country is prone to droughts (five consecutive years 2012-2016) affecting agriculture and thus rural livelihoods; annual losses for the agricultural sector were about US$ 75 mln. Irregular water supply and severe land erosion exacerbate hydrological vulnerability not only in rural areas but also in the AMSS. El Salvador is depleting its sweet water resources due to overexploitation and contamination of aquifers.
8. To combat environmental degradation and reduce social and economic vulnerabilities, a series of policy instruments have been adopted in an accelerated pace.[7] Since 2009, climate change has been mainstreamed into national policy, specifically the National Development Plan (*Plan Quinquenal de Desarrollo - PQD)* 2014-2019. *The* Objective No.7 states "*to transit towards an economy and society that are environmentally sustainable and resilient to the effects of climate change*", and identifies strategies and targets to meet this objective. Another important achievement is the creation of specialized climate change units in key Government entities including the Ministry of Environment and Natural Resources (MARN), Ministry of Finance (MH), the

con sus Anexos (LODT-AMSS) was approved.

[4] Source: UNDP Country Programme for El Salvador 2016-2020, p.3. This figure is twice the regional average and 10 times the global average.

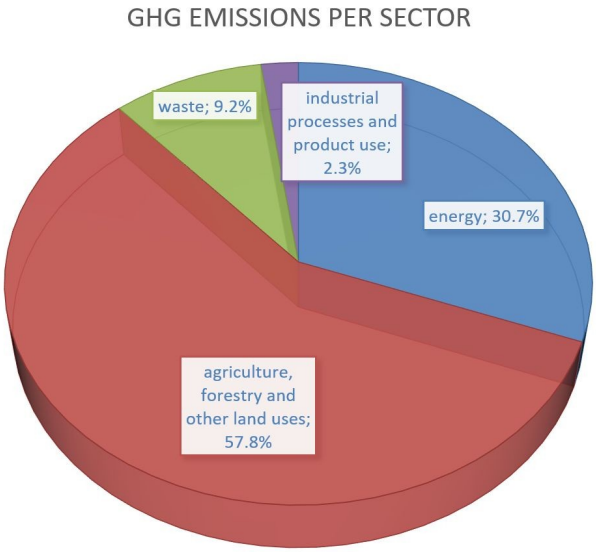
[5] EHPM = Encuesta De Hogares De Propósitos Múltiples.

[6] The tropical storms Ida (2009); Agatha, Alex y Mathew (2010); and the storm 12E (2011).

[7] For a detailed description, see Third National Communication, p.27 (2018).

Ministry of Public Works, Transport, Housing and Urban Development (MOPTVDU), the Ministry of Agriculture and Livestock (MAG), the Social Investment Fund for Local Development (FIDSL) and the Ministry of Foreign Affairs (MIREX), which have developed sectorial climate change strategies, policies and plans. Some sectors have made notable progress to align with national climate policy, especially the energy sector governed by the National Energy Council (CNE).

9. El Salvador's share in global emissions is low both in absolute terms (0.04% of total) and per capita (1.1 ton CO2e/yr). El Salvador has assumed its commitments under the UNFCCC as expressed in its NDC. The recent Third National Communication (TNC, 2018) and Biennial Update Report (BUR, 2018) indicate the following figures for national GHG emissions and absorptions by El Salvador, per sector[8]: energy (30.7%; 6,269 kton CO2e/yr); agriculture, forestry and other land uses (57.8%, 11,794 kton CO2e/yr); waste (9.2%; 1,871 kton CO2e/yr); and industrial processes and product use (2.3%; 462 kton CO2e/yr). Total emissions are 20,395 kton CO2e/yr. The GHG emissions in the energy sector are for 97.1% related to the combustion of fossil fuels for land transport and for energy generation.



10. Notwithstanding, progress is hampered by challenges concerning the institutional framework and capacities of the public administration to deliver on stated policy objectives. The PQD 2014-2019 highlights that: "a transformation of the Public Administration is needed, which requires updating of legislation as well as the processes and institutional structures of the Government". This transformation process is led by the Presidency through the Technical and Planning Secretariat (SETEPLAN) and the Secretariat for Civil Participation, Transparency and Anti-corruption (SPCTA). Strengthening of public administration competences, transparency and the generation and management of information to support policy development, are transversal themes in the PQD. The transformation would imply a certain decentralization of competences from the Central Government to the municipalities as targeted by the national Territorial Planning and Development Law (LODT, 2011), and concerns both the AMSS and the rest of the territory.
11. The Government of El Salvador (GOES) considers the pursuance of a low-emission development path as an opportunity for modernization of productive systems and services, thereby increasing competitiveness and reducing the high social and environmental externalities presently incurred into. Adaptation-based Mitigation (MbA)[9] has been adopted as the underlying principle for Government policy in the PQD 2014-2019 and the

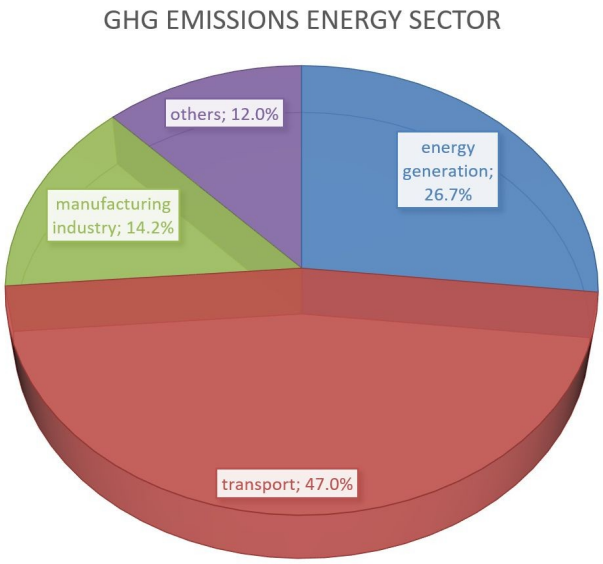
[8] BUR, p.55.

[9] MbA = Mitigación basada en la Adaptación.

National Climate Change Plan (*Plan Nacional de Cambio Climático – PNCC*). Instruments such as NAMAs and REDD+ have been identified as opportunities for attracting the necessary financial resources.

Energy sector emissions and fuel consumption

12. According to the Energy Balance 2014 of the National Energy Council (CNE), total primary energy use amounts to 128,326 TJ, of which approx. 3/10 (29.9%) is obtained from primary energy sources including geothermal (10.5%), hydropower (5.7%) and biomass (13.7%); the latter is predominantly firewood (10.1%). These sources are complemented with imports of oil derivatives (69.0%) and electricity (1.1%); as such, El Salvador imports 7/10 (70.1%) of its energy demand as secondary energy. Since 2012, when the RASA oil refinery in Acajutla was closed, the country imports all oil derivates including fuel oil (mainly for electricity generation) and gasoline (petrol) and gasoil (diesel) which are used as transport fuels. To this can be added imported LPG for residential use (alongside biomass and electricity).[10] The total value of imported fuel amounts to approx. US$ 713 million (2017), nearly 7% of the total import value over that period (US$ 10,593 million).[11]
13. The energy sector emissions by combustion of fuels (6,087 kton CO2e/yr) are distributed over the subcategories as follows: energy generation (26.7%; 1,625 kton CO2e/yr); transport (47.0%; 2,863 kton CO2e/yr); manufacturing industry (14.2%; 864.7 kton CO2e/yr); and others (12.0%; 733.3 kton CO2e/yr). The GHG emissions by the transport sector (2,863 kton CO2e/yr) are practically fully on the account of land transport. The BUR[12] highlights the need to generate more detailed data on the transport sector to enable a more accurate estimation of GHG emissions.

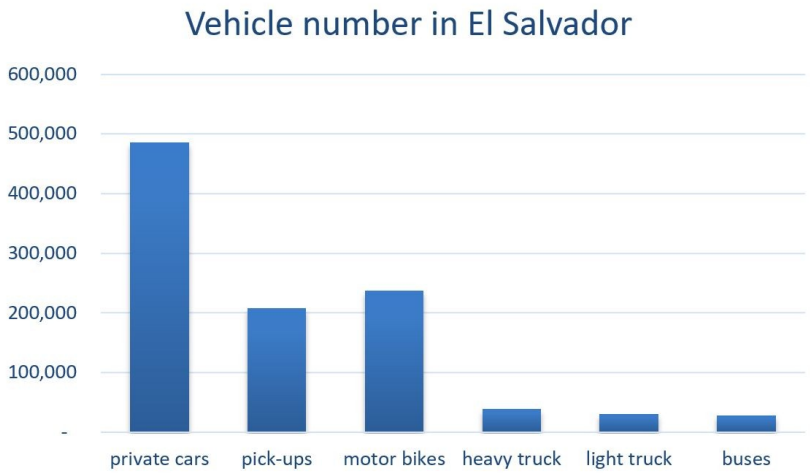


14. According to the Viceministry of Transport (VMT), the total number of vehicles in El Salvador in 2017 was 1,077,063, which represents an increase of 38% compared to 2013 (781,037). Private cars (486,037) make up the larger share (45.1%); followed by pick-ups (208,415; 19.4%); motor bikes (237,907; 22.1%); heavy truck (39,236; 3.6%); and light truck (30,290; 2.8%). Autobuses and microbuses amount to 28,577 units. An estimate of the number of vehicles circulating in the AMSS is obtained from those registered in the departments San Salvador (435,018; 40.4%) and La Libertad (149,051; 13.8%), totalling 548,069. Roughly 79.4% of all vehicles run on gasoline and 19.1% on gasoil. In the AMSS, nearly 20,000 bicycle users are registered. The GHG emissions by the transport sector (2,863 kton CO2e/yr) are practically fully on the account of land transport.

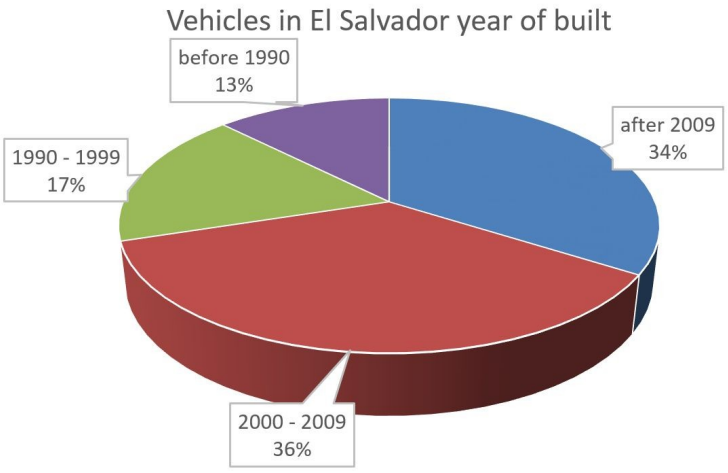
[10] BUR, p.61.

[11] Source: Banco Central de El Salvador, www.bcr.gob.sv.

[12] BUR, section 2.3.6, p.68.



15. The average age of transport vehicles in El Salvador is more than ten years. Only 32% of the total vehicle stock (350,242) was built after 2009; 35% (372,241) between 2000 and 2009; 17% (179,343) between 1990 and 1999; hence the remaining 16% is nearly 30 years old. It must be noted that the increase in private cars between 2013 and 2017 (nearly 300,000 units) includes a significant number of second-hand cars, often older than 7 years. In 2015 about 34,000 vehicles were imported second-hand from the USA, which suggests that about 35-50% of all car imports are second-hand.[13] Only 20% (5,791) of all buses and microbuses was built between 2010 and 2018; 50% (14,212) between 2000 and 2009; and 20% (5,812) between 1990 and 1999.



Electricity sector and energy efficiency

16. Structural reform processes in the 1990s reduced participation of the State in the energy sector, eventually triggering an increase in thermal power generation that led to high energy prices and a growing dependence on fuel imports. While in the 1980s El Salvador produced 77% of its electricity from hydropower, this had dropped to 50% in 2009. In 2007, the National Energy Council (CNE) was created as an instrument to realign the electricity

[13] Many of these cars can no longer circulate in the USA as they were drowned or damaged in accidents. They are sold at auctions, exported and repaired in El Salvador and thus potentially unsafe. Source: https://www.elsalvador.com/noticias/negocios/391543/ distribuidores-piden-que-no-se-importen-carros-con-mas-de-5-anos-de-antiguedad/

market with national interests.[14] The policy framework for El Salvador's energy sector is given by the National Energy Plan 2010-2024.

17. The pricing system under the regulated electricity market[15], which is based on dispatch in order of energy production costs, is expected to push obsolete thermal power plants (bunker oil) out of business. Under leadership of the CNE, the country is pushing forward a transformation of the energy matrix by adding solar PV, wind turbines and natural gas to complement existing hydro and geothermal power. El Salvador's NDC (2015) envisaged a number of actions in the energy sector, which were further articulated by CNE in 2017. For 2025, the energy sector strives at a reduction of 46% GHG emissions compared to the business-as-usual (BAU) scenario and an increase in RE-based electricity generation of 30%, compared to 2015 baseline levels. Only the energy sector has set quantified GHG reduction targets in response to the NDC.
18. Since 2014, the electricity generating capacity has grown by 393 MW, all solar and wind energy. In 2017, 56% of total production was obtained from conventional hydro power and non-conventional renewable energy technologies (RETs). In June 2016, the General Superintendency for Electricity and Telecommunications (SIGET) and the electricity distributing companies represented in DELSUR launched a public tender to purchase electricity from 170 MW non-conventional RETs under 20-year contracts starting in 2019. The response from international contractors and investors was very positive and will result in 100 MW solar and 70 MW wind power, producing electricity for approx. 280,000 households.[16]
19. Interest from the global RE market to invest in El Salvador has been very positive, which is a clear signal of trust by project owners that their assets will be secure and profitable under the present regulatory framework. In fact, the 2016 tender originally considered 50 MW wind power but this was increased to 70 MW in response to the interest from the market. As an example, the PV project *Providencia Solar* in Municipality El Rosario is currently the largest solar PV farm in Central America with a total capacity of 101 MW_p. Notable is also the interest of Salvadoran investors in the market.
20. The GEF-4 Project "Energy Efficiency in Public Buildings"[17] (2010-2014) was a pioneering experience for the country. An important result was the creation of EE committees in public entities. As of 2016, a total of 130 committees has been installed under guidance of CNE´s Energy Efficiency Directorate. The committees are now embedded into the "El Salvador saves Energy (PESAE)" programme which reunites participants from public and private institutions, universities, NGOs, international cooperation agencies and the electricity distribution companies.[18] The NEP 2010-2024 identified an estimated 6,542 public buildings in the country with a total electricity consumption of 378 GWh/yr (6.4% of total consumption). With financial support from IDB and JICA, in 2015 CNE commissioned a study to evaluate the feasibility of a trust fund for EE in the public sector and design its institutional and operational structure. Early 2017, the IDB approved additional technical assistance to strengthen the proposal.
21. In 2018, CNE prepared a Law project for this trust fund ("FIDEnergetica") for submission to the national Assembly. The fund will be administered by the public development bank BANDESAL[19], which will assume the upfront costs for project development and investment, to be reimbursed by the public entity through the monetary savings associated to the reduction in energy costs. An amendment to legislation is needed since otherwise these savings would flow back to the national budget.
22. Also in 2015, a study was carried out to prepare a Nationally Appropriate Mitigation Action (NAMA) "EE in public buildings". The proposal was presented to the NAMA Facility in 2016 for funding by the German Government and was initially selected. However, it was not included for funding in 2017. It is being redesigned for resubmission but CNE also explores other options to access funding. The FIDEnergetica will be one of the pillars of the revised NAMA and will be open to municipalities, autonomous public entities, ministries and other public institutions.

[14] Ley de Creación del Consejo Nacional de Energía (Decreto Legislativo N° 404, 2007).

[15] Mercado Regulado del Sistema.

[16] CNE Annual Report (*Rendición de Cuentas*) 2017, p.9.

[17] GEF ID 3901.

[18] PESAE - Programa El Salvador Ahorra Energía. See: http://www.pesae.org.sv/.

[19] BANDESAL = El Salvador (Banco de Desarrollo de El Salvador -

23. CNE has set a 28% reduction target for electricity consumption in public buildings and lighting for the year 2025 compared to the BAU scenario. This would be achieved by substitution of electrical appliances to be financed through FIDEnergetica. Specifically, the following measures have been identified: (i) substitution of 80% of current stock of conventional airconditioners (AC) by high-efficiency units (58.9 GWh accumulated electricity savings by 2025); and (ii) substitution of at least 80% of current T-12 units by LED (38.8 GWh savings). More details are provided in Annex C.

Energy efficiency standards for lighting and A/C appliances in El Salvador

24. In 2011, El Salvador created the Salvadoran Organisation for Technical Regulations (OSARTEC) as part of the National Quality System. The System further comprises the Salvadoran Normalization Organization (OSN), the Salvadoran Accreditation Organization (OSA) and Center for Metrological Research (CIM), under the purview of the National Quality Council (CNC). Since then, the earlier system of official (NSO) and recommended (NSR) standards has been replaced by the current Salvadoran Technical Regulation (RTS). A systemic weakness is the absence of national laboratories for product verification and testing. Hence, compliance of imports with applicable regulation is verified based on documentation provided by the supplier.
25. In January 2018 five new EE standards were approved and published by the Ministry of Economy. Due to some delay, these RTS entered into force as of 7 January 2019. The standards stipulate minimum energy performance criteria based on Mexican regulation (NOM). Most of the pre-2011 standards have now been derogated, except for compact fluorescent lamps (CFL). El Salvador further adheres to the Central American regulations under CAFTA-DR.
26. The EE standards currently in vigor, according to CNE information, are summarized in the following table.

Electric Appliances – National Regulation for minimum EE performance				
	Approval Year	Technology	RTS code and name	NOM reference for EE minimum limits
1	2017	Asynchronous three-phase electric motors	"RTS 29.01.01:15 Eficiencia Energética. Motores de Corriente Alterna, Trifásicos, de Inducción, Tipo Jaula de Ardilla, en potencia nominal de 0,746 a 373 kW. Límites, Métodos de Prueba y Etiquetado."	NOM-016-ENER-2016
2	2017	Household refrigerators and freezers	"RTS 97.01.01:15 Eficiencia Energética. Refrigeradores y Congeladores Electrodomésticos. Límites, Métodos de Prueba y Etiquetado."	NOM-015-ENER-2012
3	2017	Commercial refrigerators	"RTS 97.02.01:16 Eficiencia Energética.Equipos de Refirgeración Comercial Autocontenidos. Límites, Métodos de Prueba y Etiquetado."	NOM-022-ENER/ SCFI-2014
4	2017	Air conditioners	"RTS 23.01.03:15 Eficiencia Energética. Acondicionadores de Aire tipo Dividido, Descarga Libre y Sin Conductos de Aire. Límites, Métodos de Prueba y Etiquetado."	NOM-023-ENER-2010
5	2017	Window air conditioners	RTS 23.01.02:15 Eficiencia Energética. Acondicionadores de tipo Cuarto. Límites, Métodos de Prueba y Etiquetado	NOM-021-ENER/ SCFI-2008
6	2010[20]	Compact fluorescent lamps	NSO 29.47.01:09 "Eficiencia energética y seguridad de lámparas fluorescentes compactas integradas, requisitos de desempeño energético y etiquetado."	NOM-017-ENER/ SCFI-2008

[20] Issued by the then CONACYT, 5 March 2010. Diario Oficial Tomo No. 388 (2010).

Sustainable urban development and governance

27. Sustainable urban development requires effective governance structures for territorial planning to be in place. Urban planning in the AMSS cannot be viewed independently from the overall situation in El Salvador, which is characterized by a complex and fragmented institutional landscape. The country is in a process of maturing its institutions. Major challenges include the constrained fiscal budget, lack of professional training and career building in public administration, scattered competences and capacities over ministries and dependencies. The political polarization between the left-wing (FMLN) and right-wing (ARENA) parties undermines the efficiency of legislative processes to push forward a State reform. Governments therefore recur to Presidential Decrees with sector ministries absorbing administrative functions which further undermines public sector capacity to deliver - and a vicious circle is born.

28. The relation between governance and territorial development has been analyzed under the PPG in a separate study providing a description of involved institutions and relevant policies and legislation.[21] It was concluded that – different perhaps to other cities in Latin America pursuing integrated sustainable development with GEF support – territorial development policy does not provide a viable entry point for the AMSS and therefore another approach is chosen (see section Theory of Change). However, a basic understanding of key institutions related to territorial development and mobility in El Salvador and the AMSS in particular is essential; this is provided in the following paragraphs.
29. The most relevant institutions for sustainable territorial development in El Salvador can be classified along the levels: presidency; sector ministries; and municipalities. The presidency, its secretariats and the sector ministries form the Executive Organ. Key actors include: the Technical and Planning Secretariat (STP, or SETEPLAN) and the recently created Cabinet for Environmental Sustainability and Vulnerability[22] . At the level of sector ministries, the key institutions are: MARN, MOPTVDU[23] and CNE (the latter is not a Ministry but a Council. This level engages with civil society through the National Council for Environmental Sustainability and Vulnerability (CONASAV).[24]
30. At the municipal level there are 262 autonomous Municipal Governments bound by their territories. The Council of Mayors of the AMSS (COAMSS)[25] is the commonwealth of the 14 municipalities in the area, with its Planning Office of the AMSS (OPAMSS). Coordination of development between the AMSS and the Central Government is arranged through the Metropolitan Development Council (CODEMET)[26]. Finally, the national association of municipalities is (COMURES)[27]. This is an entity established under private Law and is not a government institution.
31. With respect to consultation with civil society, mandatory mechanisms are foreseen as part of policy making and legislative processes. Extensive consultation has taken place for example, in the context of the current National Five-Year Development Plan (PQD, 2015), the National Climate Change Plan (PNCC, 2012), and the National Plan for Territorial Planning and Development (PNODT, 2004). The PNODT identified a need to strengthen associative structures between municipalities and the establishment of micro-regional offices with technical capacity to implement competences and services on behalf of their municipalities.[28] It envisages a National System for Territorial Information (SNIT) to reduce the fragmentation of data collection and management among various public and private actors, ensure data quality. In 2011, it was approved under the Territorial Planning and Development Law (LODT). However, the Plan is very ambitious taking into account the limited budget and human resources.

Territorial development in the San Salvador Metropolitan Area (AMSS)

32. The approval of the LDOT-AMSS by the Assembly integrated the COAMSS, OPAMSS and CODEMET into national legislation (1995). The same Law stipulated the preparation of a Master Plan for the AMSS (the *Esquema Director*) which was eventually delivered in 2016. The delay of 22 years demonstrates the slow pace of institutional reforms and the lack of resources and political prioritization. Notwithstanding, a first Integrated Urban Development Plan of the AMSS[29] was prepared by the Ministry of Public Works (MOP) in 1967-1969, known as METROPLAN 80. The plan proposed a zoning of the city and a primary road system. In 1988, another plan was produced under the name METROPLAN 2000, which featured the decentralization of central

[21] See Report: Gap Analysis of Public Administration Capacities to Implement Low-emission Urban Development Programmes – Low-Emission Development Governance, by R.Rijs, August 2018 (UNDP El Salvador, funded by GEF under PPG) - ANNEX M.

[22] Gabinete de Sustentabilidad Ambiental y Vulnerabilidad.

[23] The Ministry of Public Works, Transport, Housing and Urban Development.

[24] CONASAV = Consejo Nacional de Sustentabilidad Ambiental y Vulnerabilidad

[25] COAMSS = Consejo de Alcaldes del Área Metropolitana de San Salvador.

[26] CODEMET = Consejo de Desarrollo Metropolitano.

[27] COMURES = Corporación de Municipalidades de la República de El Salvador.

[28] The role of the regional offices would be comparable to OPAMSS.

[29] Plan Integral de Desarrollo Urbano del AMSS

Government decisions in favor of the municipalities. The Plan expanded the area defined as the AMSS to 13 municipalities and pursued a controlled and balanced development in the area.[30]

33. The present Master Plan is the first step towards the Metropolitan Territorial Development and Planning Plan of the AMSS[31], which further involves sectoral public investment plans. Development of the Master Plan took place between 2013-2016 with financial support from international cooperation. It identifies categories of land uses and open areas and assesses the functionality of urban systems on a macro-scale. An analysis of bio-physical conditions includes threats and risk factors such as volcanic activity, resulting in a mapping in terms of suitability for urbanization. The Plan also describes socio-economic conditions per municipality.
34. Future development of the AMSS is described through three scenarios: (1) no planning intervention (*escenario tendencional*); (2) full compliance with the vision of the Master Plan (*escenario óptimo*); and (3) partial implementation of the Master Plan (*escenario intermedio*). The latter is arguably the most realistic and provides a basis for evaluating costs and benefits of policy interventions. The Master Plan identifies a set of indicators for monitoring the status and progress of four urban sub-systems: population; environment; settlements; and infrastructure. The Plan identifies a series of problems per municipality as well as opportunities and potential solutions. For 2030, it envisions "a sustainable, inclusive, competitive and resilient city in the Central American context, with a polycentric configuration and in the process towards densification and structuring around a grid of public spaces; and a new multimodal transport system offering opportunities to all under a financing model that is sound, progressive and diverse."[32]
35. In December 2017, the Initial Adaptation Plan for the AMSS was delivered for COAMSS/OPAMSS and CODEMET[33]; this responds to a commitment made in El Salvador's NDC. The Plan observes the lack of input data, obsolete information, inconsistencies in forecasts and estimates, changes in methodologies and reporting methods which impedes an adequate comparison between earlier situations and the present status. It also stresses the lack of defined competences for rain water management in the AMSS and in El Salvador in general. The Plan reiterates the importance of developing an adequate institutional structure for its implementation. It also identifies the opportunity to take benefit from the technical working groups established under the CODEMET.

Mobility and public transport in the AMSS

36. In 2010 COAMSS approved its urban mobility policy paper[34] which seeks to incorporate urban mobility into the environmental, social, economic, cultural and political domains of the AMSS municipalities. The policy aims to establish the principles underpinning a vision on sustainable mobility, essentially from the perspective to reduce displacements and substitute polluting, congesting modalities by socially and environmentally more amicable ones. It acknowledges that a successful approach goes beyond mobility alone as it involves public policies from other sectors at the central and local government levels. The policy acknowledges that mobility is intertwined with land use.
37. The COAMSS policy provides an overview of the status of public and private transport in the AMSS at that time, as well as the applicable legal instruments. The document ascertains that "*in general terms, the legal competence for hard mobility is with the central Government while for soft mobility it is with the municipalities*".[35] It observes that the municipalities must obtain permission from VMT for the implementation of civil works and equipment on public roads and coordinate the location of bus and cargo terminals, bus stops and returns. In practice, this greatly reduces the capacity of municipalities to pursue a mobility policy according to their own insights.

[30] Source: Plan Inicial de Adaptación del AMSS, p.100.

[31] Plan Metropolitano de Desarrollo y Ordenamiento Territorial del AMSS.

[32] Esquema Director AMSS, Resumen Ejecutivo, COAMSS/OPAMSS, p.58 (2016).

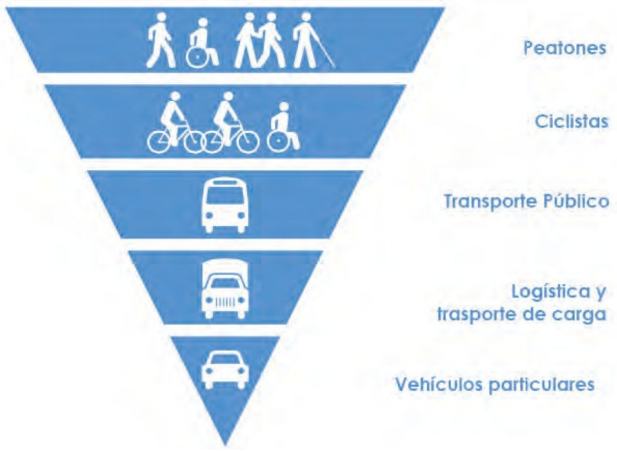
[33] Plan inicial de Adaptación del AMSS, Carlos Tucci, Lilliana Arrieta Quesada, e.a., for COAMSS/OPAMSS, MARN, CODEMET and UNDP, December 2017.

[34] Política de Movilidad Urbana – Esquema de Coherencia Territorial, COAMSS/OPAMSS, 21 October 2010.

[35] Ibidem, p.7.

38. Progress to implement the COAMSS policy has been - at best - modest. A key barrier is that the policy relies on a transfer of competences in the realm of mobility from VMT to COAMSS. Specifically, the policy proposes the creation of a Municipal Transport Authority. Arguably, some activities can be more effectively handled by a lower authority[36] but political back-up from the Executive Organ and/or the Assembly is crucial. A political dialogue on this issue has not crystallized (or has not even begun).[37] The modest results so far should also be viewed in relation to the timeframe set in the policy, which is not realistic.

39. The AMSS Master Plan (2016) also embraces the concept of sustainable mobility involving more compact urbanization and less dependence on motorized traffic. It is acknowledged that VMT is traditionally focused on motorized transport and the development of physical infrastructure rather than on traffic management to optimize utilization of infrastructure and reduce fuel consumption and displacement times, and other externalities. The concepts of non-motorized modalities (bicycle use, pedestrian routes) are not captured in Law, and regulation of related infrastructure and traffic management measures is lacking. Traffic regulation of bike riders is rather paternalistic and moralising, literally reducing their appearance to the margins of public space. The current situation can best be described by the mobility pyramid, having VMT and car mobility at the apex and OPAMSS and urban strollers at the bottom side, which is inverse to the situation envisaged by COAMSS in the Master Plan.



40. Clearly, besides a change in perception by society, changes in the institutional and legal framework for urban mobility involve complex political processes. The creation of a municipal Transportation Authority can be a solution for the AMSS but other options are equally possible.

Public transport in the AMSS

41. Public bus transport in the AMSS involves 162 routes and 42% of the total public transport fleet in the country.[38] An instruction has been in place for VMT since 2011 to reorganize the bus routes in the AMSS and make these more efficient, but the process has been slow and politically sensitive. The poor quality of public transport is driven by the concessional model and the tariff system, which are poorly implemented and produce perverse stimuli that foster chaos and insecurity. Since interpretation of the Public Procurement and Contracting Law

[36] Many countries try to align public functions with the subsidiarity principle, which implies that they should be organized at the lowest possible government level when feasible. However, other perspectives also exist: a more centralized organization can facilitate standardization and promote economies of scale.

[37] There is no legislation in place that provides a basis for such transfer, hence a reform of the Regulation of the Executive Organ (Art. 45) may be required, among others.

[38] Source: SITRAMSS – Estimación preliminar acerca del potencial de reducción de emisiones de gases de efecto invernadero y otros contaminantes atmosféricos, Juan Pablo Domínguez Miranda, Pablo Hernández Panameño, Wester del Cid Ayala, Consultores Biosistemas S.A. de C.V., January 2015, p.19 ff.

(LACAP, 2000)[39] seems to be ambiguous, VMT has awarded the bus routes to prospective operators as a kind of license. Since the licenses are not exclusive several operators concur at the same bus stops competing for the same passengers. The lack of a fixed and adequate income for the drivers generates a "struggle for life" resulting in poor service, insecure operation and infractions to traffic regulation.[40]

42. In June 2011, VMT issued a first tender for bus routes in rural parts of the country[41] in response to Art. 47 of the *Ley General de Transporte Terrestre, Tránsito y Seguridad Vial* and the National Procurement Law (LACAP) Art. 49, 133 and following. However, the system of concessions was not further implemented. Obviously, the licenses have created vested interests which are difficult to revert as short-term political costs are high. Although little information is available, experiences worldwide indicate that a licensing system is prone to corruption and fosters informality due to trading in licenses and concentrating thereof in few hands creating monopolies.
43. For political reasons, the tariffs are set by VMT below the level required for profitable exploitation (US$ 0.20 instead of somewhere in the range of US$0.30-0.60). A compensation was introduced in 2007 by creating a subsidy of US$400 per month per bus (US$ 200 for a microbus) under a temporary Law.[42] In 2015, this was changed into a prime paid per transported passenger (US$ct4 for buses, US$ct 2 for microbuses), to be applied only to authorized buses in the AMSS. These subsidies are paid through the COTRANS[43], a levy of US$ 0.10 per gallon gasoline and gasoil purchased in the market.
44. The reportedly low revenues by bus owners are compensated by demanding high daily returns from their drivers and saving on bus maintenance. A uniform electronic payment system for public transport was designed by national company SIPAGO to facilitate accountability and prepare the SITRAMSS integrated bus system (see below). In spite of all security benefits by reducing the amounts of cash money in the buses, the system was not accepted by the transport sector and VMT could not enforce its implementation. The business model is driven by intransparency and favors the use of obsolete technology, which only prospers if regulation and quality standards are not enforced.[44] In the absence of an autonomous oversight agency, enforcement of governance by VMT comes at a political cost.

SITRAMSS Bus Rapid Transit

45. The Bus Rapid Transit (BRT) is a mass transport modality that has been implemented in many cities worldwide. Examples in Latin America include Curitiba, Bogotá, Guatemala and Mexico City. In El Salvador, the BRT is called the AMSS Integrated Transport System (SITRAMSS)[45]. Previously, a Master Plan was made that identified seven main corridors running east-west and north-south along the city's main roads. The first corridor for the BRT would run from San Martin in the east to Santa Tecla in the west; for the future (2025) a second Corridor is planned running north-south.
46. A first stretch (Phase I) of 6.4 km was implemented between 2013 and 2015 running from the *Hospital Médico-Quirúrgico* in San Salvador (W) to Soyapango (E). This section counts with a dedicated bus lane and elevated platforms congruent with the technical requirements for a BRT. The infrastructure was financed through a loan from IDB.[46] VMT, who oversees the project, decided to expand the route 3.1 km further west until the roundabout *Divino Salvador del Mundo* which is a landmark in the city; the buses use the normal road infrastructure over this stretch.

[39] LACAP = Ley de Adquisiciones y Contrataciones de la Administración Pública.

[40] As confirmed by focal groups convoked in the framework of the study "Prevención del crimen en el transporte público en El Salvador", Fundación Salvadoreña para el Desarrollo Económico y Social (FUSADES), Project SolucionES under USAID contract AID-519-A-12-00003, June 2015.

[41] Source: VMT press release "VMT licita por primera vez concesiones de transporte colectivo", 29 June 2011, www.vmt.gob.sv.

[42] Ley transitoria para la estabilización de las tarifas del servicio público de transporte colectivo de pasajeros, December 2007.

[43] Contribución Especial para la Estabilización de las Tarifas del Servicio Público de Transporte Colectivo de Pasajeros (COTRANS).

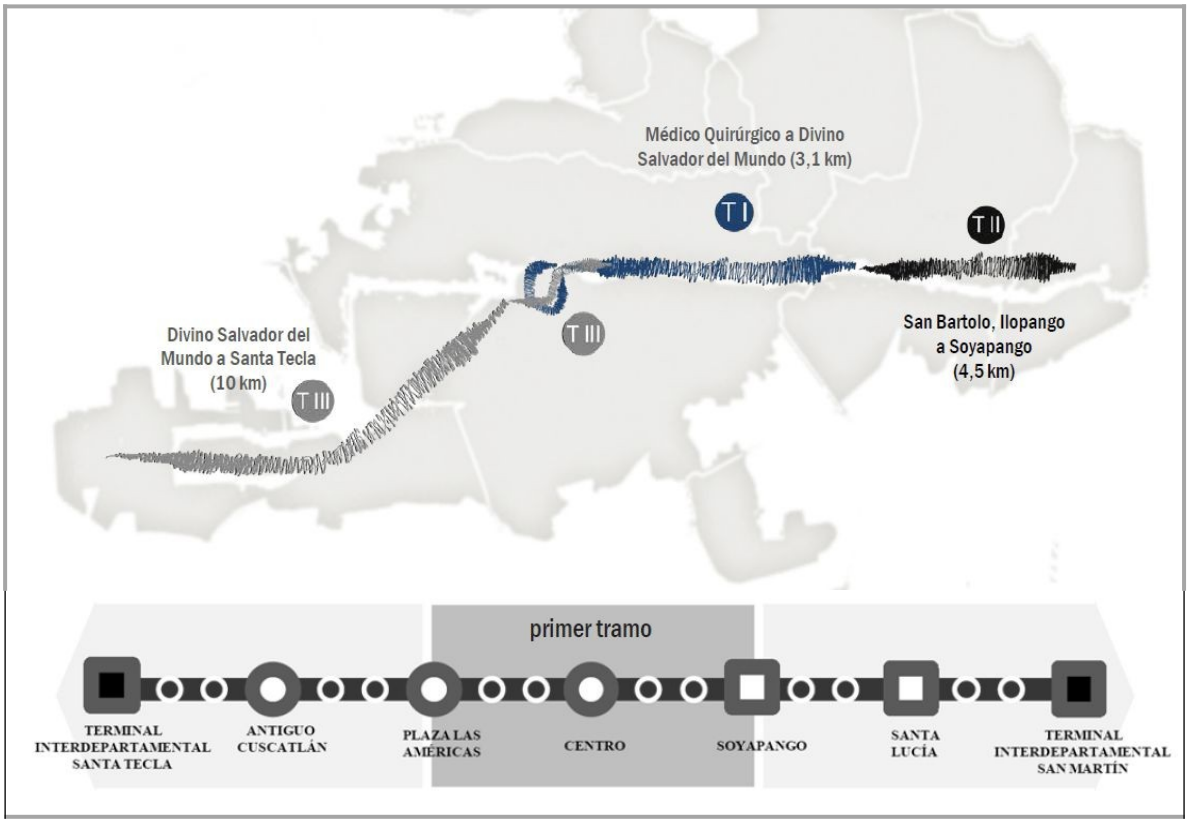
[44] Specific regulation and quality criteria exist though. Reglamento general de transporte terrestre Art. 90, 91. Ley de transporte terrestre y seguridad vial, Art 31. Reglamento de transporte terrestre, tránsito y seguridad vial, Art. 59, 65, 70, 73, 75, 80, 8182, 98, 102, 104, 188, 244.

[45] SITRAMSS - Sistema de Transporte Integrado del Área Metropolitana de San Salvador.

[46] IDB loan ES-L1050/2752/OC-ES "Programa de Transporte del Área Metropolitana de San Salvador" to an amount of US$45 mln.

47. The SITRAMSS currently operates 21 short buses and 16 articulated units operated with diesel engines under EURO-3 standard. The envisaged number of buses is 151 short buses and 57 articulated ones on the east-west Corridor I. The number of passengers has been estimated at 523,000 per day along the entire Corridor, which is about 40% of current total public transport demand in the AMSS. The expected passenger rate on the existing Phase 1 is about 75,000 per day. Where previously traffic speed was 5-11 km/h average, the BRT would reach 20 km/h including the halting times at the bus stops.

48. Implementation suffered a severe setback in May 2017 however when a group of citizens successfully recurred to the Constitutional Court claiming that the contract between VMT and SITRAMSS was in breach with Constitution Art. 120, as it should have been subject to the Assembly prior approval. The Court has not issued a verdict since but issued a provision (*Medida Cautelar*) ordering the preferential lane to be opened for all traffic on 11 May 2017. Since then, the advanced BRT units run together with the common traffic, even competing with the cheaper conventional bus routes which could not be taken out of service. Passenger utilization rates have fallen far below the estimations and the anticipated social, economic and environmental benefits are not obtained.[47] Negotiations between SITRAMSS and the BNDES[48] for financing of additional bus units have been suspended.



49. The current status is not only a product of incompliance with public administration laws (or flexibility in their interpretation). An underlying problem is that the BRT has been used as an argument for political campaigning. This put pressure on the Government to respond to the high expectations created, undermining a proper due diligence process. The project was pushed forward by VMT which assumed overall political and technical responsibility, leaving many open ends including: (i) full participation of existing transport companies in SITRAMSS; (ii) careful design of itineraries and implementation of a traffic plan; (iii) separation of supervision

[47] Quantitative data have not been made available but considering the capacity drop due to the reduced transit speed (of the order of 8 km/h) daily passenger rates are probably of the order of 20,000-25,000 persons per day (educated guess).

[48] BNDES = Brazilian Development Bank.

and operation of SITRAMSS according to a transparent business model; (iv) utilization of public infrastructure and funding modality for vehicles; and (v) reorganization of conventional bus routes.

50. With all vested interests in the transport sector in the AMSS, persuasion of stakeholders is critical to achieve political support in the Assembly as well as in the municipalities. Without such support in place SITRAMSS became an easy target for opponents, be it for legal-technical or for political reasons. In this context, one of the key weaknesses of the transport sector in El Salvador should be recalled, i.e. that it is managed by a political entity (VMT) and not a technical public body. Notwithstanding these deficiencies, the perception of the BRT among the general public remains quite positive; early users valued the shorter journeys and the impact thereof on their lives.[49]

Barrier analysis

51. Sustainable, low-emission urban development in the AMSS is hampered by a series of barriers. These barriers include: (a) policy and regulation: weaknesses and voids in the institutional and legal framework, including inadequate professional education of public administration offices; lack of quality standards for public lighting and mobility and poor enforcement thereof; (b) technology: inadequate in-country know-how, methodologies and tools for evaluating mobility scenarios and designing low-emission solutions; (c) delivery models: lack of adequate business models for investment and operation of public lighting and public transport including SITRAMSS; (d) finance: fiscal and legal constraints to leverage public and private investment capital, and a lack of experience with public-private partnerships; and (e) information: inexistent or inaccessible data for baseline definition, design of interventions and monitoring of their impact.
52. Overarching barriers are the lack of political consensus and systemic governance barriers to push forward territorial development and sustainable mobility strategies. The division of mandates between sector ministries, and between the central Government and the municipalities, gives rise to conflicting and/or overlapping competences making urban development planning and enforcement thereof complex and ineffective. With respect to energy efficiency by municipalities, the primary barrier is the lack of effective outreach mechanisms and access to finance.
53. This situation is exacerbated by systemic barriers and root causes including the small size of El Salvador with emigration of qualified people; frequent occurrence of natural disaster, causing setbacks in the development process; and social insecurity (violence). There is no tradition of monitoring the impact of economic, social and environmental policies and interventions, and methodologies and data to this purpose are, at best, incipient. Moreover, information and statistical data are often incomplete, outdated, or based on unclear methodologies. Technical, managerial and institutional capacities need to be improved.
54. For a detailed analysis of the indicated barriers, reference is made to the GEF CEO Endorsement Request document of the Project.

Development problem

55. The development problem addressed by the Project is formulated as follows: “The transition towards a low-emission development path in the San Salvador Metropolitan Area (AMSS) is hampered by the weak regulatory and institutional framework for urban planning, insufficient or obsolete data for urban planning, a lack of proven, transparent business models for public services (including public transport), and the constrained technical and financial capacities of municipalities.”
56. Overarching barriers include the fragmented and weak institutional framework impeding effective governance of the public transport sector in the AMSS, the incipient process towards more inclusive and sustainable territorial development, and the complex and polarized political climate in the country.

[49] Source: El último día de exclusividad del SITRAMSS, La Prensa Gráfica, May 11, 2017.

III. STRATEGY

Development objective

57. The goal of the Project has been formulated as follows: "To promote a low-emission urban development path in the Metropolitan Area of San Salvador (AMSS)." The immediate (development) objective is: "To enhance national competences in the field of low-emission urban planning by addressing regulatory voids for urban mobility and public lighting, by fostering in-country capacities and skills, and improving coordination between Government stakeholders and lower authorities in the AMSS, , and develop a first batch of pilot projects for learning and demonstration of benefits and upscaling potential."
58. A specific, related objective is to establish the necessary conditions for enabling the operation of Phase I of the Bus Rapid Transit (BRT) in San Salvador (SITRAMSS) to its full potential.

Approach and theory of change

59. During the PPG it became clear that some critical assumptions underpinning the original project idea (PIF) were not fulfilled. Specifically, the mechanisms for planning of the AMSS territory are still very weak due to a range of institutional and capacity barriers which are largely systemic. In fact, effective governance is one of the main challenges pursued in the National Development Plan 2014-2019. Several efforts into this direction are being deployed, including the Territorial Planning and Development Law (LODT, 2011) and the AMSS Master Plan (2016). Their slow progress demonstrates that implementation of an effective governance system for territorial planning in El Salvador and the AMSS is a matter of decades and by far outmatches the scope and time horizon of a GEF Project.
60. The PPG team concluded that territorial development policy does not provide a useful entry point for the Project to deliver sustainable results. Moreover, the Project would endeavor into a variety of development problems including territorial development, public transport, governance, and energy efficiency. It would involve a large number of public sector stakeholders, several of which showed little problem ownership. Bringing together these stakeholders in a productive manner requires a level of internal coordination that is not currently present. Therefore, an alternative perspective has been taken, departing from the National Energy Plan 2010-2024, with the overarching objective of fuel conservation and improving the national commercial balance. Ownership of this problem is assigned to CNE, which reunites the key stakeholders in its Executive Board.
61. Work by the PPG team (including the conceptual design of the "RAPS") and engagement with national stakeholders led to the conclusion that many basic elements for building modern mobility solutions are non-existent in El Salvador. There is little experience in the country, both in theory and in practice; solutions that work in other countries are copied but conditions are not in place. This conclusion applies to simple solutions as painting of bike lanes and the introduction of 30-km zones in urban areas (not foreseen in regulation), but similarly to the definition of a segregated BRT lane for a SITRAMSS. Therefore, know-how on sustainable urban planning and mobility needs to be drawn into the country, for example through the GEF Global Platform for Sustainable Cities (GPSC). Moreover, experience should first be gained at a small scale to avoid financially costly setbacks as occurred with SITRAMSS.
62. Concerning energy efficiency in public buildings and lighting, the PPG found that green procurement is essentially covered already under the Public Procurement Law (LACAP). The key problem is the lack of a verification mechanism to purview procurement by public entities outside the Central Government. (It is recalled that the municipalities are politically and financially autonomous - with limited financial means, though.) By consequence, it is not straightforward to continue the approach followed under the GEF-4 EEPB Project to the municipalities.[50] Enforcement of technical standards and EE criteria must therefore rely on indirect action, including: (1) awareness raising and creation of municipal energy committees and (2) a due diligence process for municipal EE projects seeking external funding, which is a key element of the FIDEnergetica proposal.

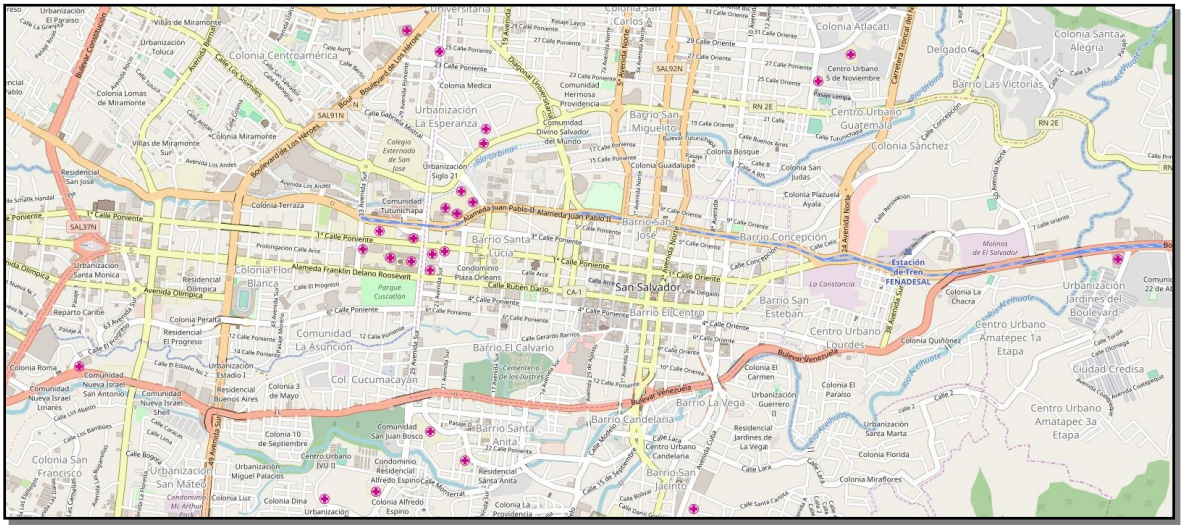
[50] The GEF Project Energy Efficiency in Public Buildings (GEF ID 3901) essentially targeted buildings under direct responsibility of the Central Government.

In any case, capacity building within municipalities is critical developing a pipeline of mature EE projects in the AMSS.

63. The National Quality System exhibits some weaknesses including the absence of laboratories for product verification and testing. Hence, compliance of imports with applicable regulation is only verified on paper. Importantly, El Salvador presently has no technical standards and EE criteria in place for public lighting products. Moreover, there are no formal technical guidelines for the design of such systems. This especially affects the functionality of public lighting managed by municipal authorities. The GEF Project can make an important contribution to fill this gap by developing such standards and promoting these among the AMSS municipalities.
64. The Project design deviates somewhat in scope from the original proposal (PIF). For a comparison at output-level, reference is made to the CEO Endorsement Request of the Project. Most important are: (i) The recognition that BRT expansion will unlikely occur under the baseline; the investment in additional bus units cannot be confirmed at this stage (co-finance). Instead, the Project aims at creating the conditions for SITRAMSS to function as designed, specifically by pursuing a SITRAMSS Law and strengthening the business model; (ii) The lack of regulation, and in-country know-how and expertise with sustainable urban planning, specifically low-emission mobility concepts and standards for public lighting systems. Transfer of know-how through international partners (GPSC, mobility experts, technical standards for lighting) is therefore one of the cornerstones of the current Project.
65. The impact of low-emission solutions will be demonstrated through a number of representative pilots in four selected municipalities. These pilots include pedestrian routes and zones, bicycle lanes with bike rental services, traffic management (including one-way roads, traffic signs and lights, traffic access regulation, and adjustments to road design and capacity). Indicators for gauging local and overall impact will be developed and applied to each pilot for reporting and learning. It is envisioned that all pilots will facilitate access to the BRT SITRAMSS and as a result, increase its utilization rate and achieve associated GHG emission reductions (per passenger-km) compared to the current baseline. Energy efficiency in municipal buildings and street lighting will be promoted by facilitating access to finance and by technical assistance for developing a project portfolio.

Geographical scope

66. The Project will cover the Metropolitan Area of San Salvador in the Republic of San Salvador, Central America, with map coordinates (13.699N,-89.191W). Project activities will focus on the municipalities Santa Tecla, Antiguo Cuscatlan, San Salvador, and Soyapango (from west to east), covering a rectangular area of about 17km long between (13.663N,-89.306W) and (13.731N,-89.120W). The length of the current SITRAMSS bus route is 7.8km, from the roundabout Divino Salvador del Mundo in San Salvador Municipality to the Shopping Center Soyapango (in Soyapango Municipality).


San Salvador
Barrio Santa Lucía
Barrio El Calvario
Barrio La Vega
Barrio Candelaria
Barrio San Jacinto
Barrio San Esteban
Barrio Concepción
Barrio San José
Barrio El Centro
Barrio San Miguelito
Barrio Santa Anita
Colonia Centroamérica
Colonia Flor Blanca
Colonia Atlacatl
Colonia Sánchez
Colonia Guadalupe
Colonia Bosque
Colonia Medica
Colonia Terraza
Colonia Peralta
Colonia Roma
Colonia Miramonte
Colonia Lomas de Miramonte
Colonia Las Victorias
Colonia Santa Alegría
Colonia Plazuela Ayala
Colonia San Judas
Colonia Quiñónez
Colonia El Paraíso
Colonia Florida
Colonia Miraflores
Colonia El Carmen
Colonia El Paraíso
Colonia La Chacra
Colonia 10 de Septiembre
Colonia 3 de Mayo
Colonia Alfredo Espino
Colonia Dina
Colonia Luz
Colonia La Providencia
Urbanización La Esperanza
Urbanización Siglo 21
Urbanización Santa Mónica
Urbanización Toluca
Urbanización Estadio I
Urbanización Miguel Palacios
Urbanización Guerrero II
Urbanización Santa Marta
Urbanización Jardines del Boulevard
Urbanización San Mateo
Urbanización
Comunidad La Asunción
Comunidad Tutunichapa
Comunidad Divino Salvador del Mundo
Comunidad Hermosa Providencia
Comunidad San Juan Bosco
Comunidad Nueva Israel Shell
Comunidad Nueva Israel San Antonio
Comunidad Nueva Israel Linares
Residencial Olímpica
Residencial El Progreso
Residencial Buenos Aires
Residencial Jardines de La Vega
Residencial Santa Anita
Residencial San José
Condominio Plaza Orleans
Condominio Residencial Alfredo Espino
Centro Urbano 5 de Noviembre
Centro Urbano Guatemala
Centro Urbano Lourdes
Centro Urbano Amatepec 1a Etapa
Centro Urbano Amatepec 3a Etapa
Centro Urbano Candelaria
Ciudad Credisa
Col. Cucumacayan
Colegio Externado de San José
Cementerio de los Ilustres
Parque Cuscatlán
Estación de Tren FENADESAL
Molinos de El Salvador
Lo Constancio
Alameda Juan Pablo II
Alameda Franklin Delano Roosevelt
Boulevard Venezuela
Boulevard de Los Héroes
Boulevard Constitución
Avenida Olímpica
Avenida Los Andes
Avenida Bernal
Calle Gerardo Barrios
Calle Rubén Darío
Calle Arce
Calle Concepción
Calle Delgado
Calle Lara
Calle Monserrat
Calle Los Sobrinos
Calle Los Abetos
Calle Los Almendros
Calle Los Bambues
Calle Los Cipreses
Calle Los Pinos
Calle Los Rosales
Calle Los Sauces
Calle Los Sisimiles
Calle Los Tilos
Calle Los Tulipanes
Calle Los Tulipanes
Calle Los Tulipanes

among others (output 1.1.1) It will further prepare a Law Project to strengthen the legal basis for SITRRAMSS and to facilitate financing of EE by municipalities and autonomous public entities (output 1.1.2).

69. This component further aims to address the lack of technical regulation in the field of urban mobility (1.1.3) and increase awareness and proficiency of public officers and other stakeholders to apply modern urban planning concepts (1.1.4). In the understanding that private sector agents (both large investors and SME) can play a larger role to deliver public transport and lighting services, business models and contracting modalities need to be developed (1.1.5). It is highlighted that El Salvador as yet has little experience with mobilization of the private sector, with the exception of energy generation and distribution under leadership of CNE. Finally, this component aims to streamline financing for low-emission urban development in the AMSS seeking up-scaling of investment and strengthening the role of intermediaries and project developers (1.1.6).
70. In the End-of-Project situation (EOP), the following results are expected: (i) SITRAMSS Law project and FIDEnergetica Law are presented and approved by the national Assembly; (ii) at least one financing mechanism for integrated urban development endorsed by the Government; and (iii) 160 individuals (staff) from municipalities and central Government entities trained on low-emission urban planning (50% male, 50% female).

Output 1.1.1. Facilitation of a white paper towards an effective institutional framework for planning and financing of integrated low-emission urban development in El Salvador (GEF US$ 130,000; co-finance US$ 900,000).

71. This output aims to facilitate a dialogue between key entities in the Executive Organ (SETEPLAN, MH, CNE, MOPTVTDU) with the purpose to clarify roles and competences in the field of urban development in the AMSS. It will take benefit from existing coordinating platforms, specifically the Cabinet for Environmental Sustainability and Vulnerability chaired by MARN, and the Metropolitan Development Council (CODEMET).[53] Similarly, CNE's Executive Board provides a useful entry point for consensus-building in alignment with the National Energy Plan 2010-2024.
72. Being El Salvador in a process of articulating the roles of the central Government and the lower authorities, different perspectives exist on how competences should be defined in the future. Sustainable urban planning (including territorial planning, mobility infrastructure, and climate change mitigation and adaptation actions) requires an integrated perspective based on social, economic and environmental benefits which goes beyond the traditional sectorial perspective. Hence authorities will need to re-think institutional competences and develop appropriate methodologies and technical regulation for addressing urban development situations. For example, technical standards for road design and public lighting by MOP/VMT need to be adapted to urban environments, integrated into the national quality system (SNC) and made available to municipalities.
73. The Project aims to foster a convergent process towards a more effective institutional and regulatory framework through the delivery of a white paper endorsed by key Government stakeholders. This output will support a series of high-level meetings to push forward a common agenda on integrated urban planning, including mobility, the specific situation of SITRAMSS, and public lighting. Presentations on specific issues will be prepared and recommended solutions discussed. The Project's Technical Advisor will thereby act as the key resource person. National and/or international experts will be drawn into the process as needed.
74. GEF funds under this output will be used for: (i) one long-term consultancy for the Project Technical Advisor (international consultant) to provide technical advice and facilitate the dialogue process; (ii) two Technical Working Group Experts (national consultants) to be assigned to VMT and OPAMSS as focal persons; and (iii) one or more contracts with specialized firms or individuals to provide inputs and advice on sector governance.

Output 1.1.2. Design and advocacy for the adoption of a SITRAMSS[54] and FIDEnergetica Law by the National Assembly[55] (GEF US$ 80,000; co-finance US$ 700,000).

[53] CODEMET, led by COAMSS, was formally established under the AMSS Law in 1995 but was only recently activated as a platform for coordination between the central Government and the AMSS.

[54] The integrated Bus Rapid Transit System in the AMSS.

[55] To be approved by the national Assembly, i.e. a Decreto Legislativo.

75. Implementation of the SITRAMSS BRT currently relies on VMT compromising its role as the supervisory organ ensuring compliance with quality standards. While direct VMT involvement initially sped up SITRAMSS implementation, it became clear that the legal basis is insufficient for effective deployment of the system and to attract investment for further expansion. In this respect, El Salvador can draw on successful governance models for BRTs in other cities in the region (such as Metrobus in Mexico City).
76. A key aspect of a BRT is the use of privileged lanes; however, these proved controversial and were suspended by the Constitutional Court in May 2017. Currently, SITRAMSS cannot operate as intended and does not deliver the anticipated social, economic and environmental benefits. The legality of other aspects of SITRAMSS, notably the nature of the contract with the operator, was also challenged. Given the scope and scale of the SITRAMSS for the country, ample political support is a prerequisite for ensuring its long-term development. The most appropriate way to ensure such support is by anchoring the project into Law to be passed by the national Assembly (Decreto Legislativo).
77. Aspects of SITRAMSS to be covered in this Law include: (i) ownership of SITRAMSS as an autonomous, public or mixed-capital company; (ii) contracting or concession modality; (iii) enforcement of quality standards; (iv) roles of public and private partners for ongoing investment in fixed and rolling infrastructure; (v) responsibilities of the bus operator; (vi) cost transparency and incentives for service expansion; (vii) pricing level for customers; (viii) operational safety, insurances and other liabilities; and (ix) law enforcement including use of security personnel.
78. In May 2018, CNE prepared a Law Project for establishing a EE trust fund for public sector entities (FIDEnergetica), to be presented to the national Assembly. UNDP CO is partner in this process providing advocacy. FIDEenergetica enables capturing the revenues from energy savings by municipalities into an independent fund that allows paying back the investment capital to the lender. Since under existing law these savings would flow back into the fiscal budget, specific legislation is necessary. The proposed scheme, which involves BANDESAL, is outlined in Annex P. Since the approval and funding of FIDEnergetica is subject to political and external factors, its timeframe is still uncertain. The GEF project will make an effort through advocacy and advice, with institutional backup from UNDP.
79. GEF support to this output will cover the costs of: (i) hiring of Project TA for advocacy and liaison; (ii) hiring of two Technical Working Group Experts to act as VMT and OPAMSS focal persons; and (iii) hiring of one or more specialized firms for legal advice and finalization of Law Proposals for SITRAMSS and FIDEnergetica.

Output 1.1.3. Integration of low-emission urban mobility concepts and related technical standards into applicable regulation (GEF US$ 90,000; co-finance US$ 700,000).

80. Existing definitions and standards as used by MOP/VMT and FOVIAL are less appropriate from the perspective of urban planning and mobility. This output aims to complement current regulation, specifically to define and integrate concepts related to non-motorized mobility including pedestrian and bicycle routes; mixed utilization of infrastructure; traffic control measures such as low-speed zones; horizonal and vertical traffic signalization; temporary access for vehicles; classification of parking space; and more. Current regulation departs from a paradigm of uniform and unrestricted access for private cars rather than treating other mobility forms equally or - more appropriate in dense urban space – discouraging private vehicle use. Work during the PPG on the “RAPS” led to the insight that even simple improvements for non-motorized mobility (such as differentiated coloring of pedestrian lanes) lack regulation impeding VMT and municipalities to experiment with such modalities. Once regulation is updated, technical standards need to be developed or can adapted from other countries.
81. it is observed that space for “greenfield development”, where a BRT can take full advantage of speed and scale, is not commonplace in the AMSS. Physical constraints include: (i) limited road width with few alternative routes; (ii) obstacles such as trees, curves, slopes and complex road crossings; (iii) the widespread presence of informal activity on or along streets and roads; and (iv) poorly managed and enforced conventional bus routes generating traffic congestion. This situation demands SITRAMSS to be carefully adjusted to the local situation and anchored in municipal traffic management plans. The latter inevitably involve engagement and negotiation with stakeholders at the level of municipalities and neighborhoods.

82. This output will establish and facilitate a technical working group (TWG) consisting of VMT[56], OPAMSS, CNE and other stakeholders with the purpose to assess current regulation, standards and approaches for urban mobility in El Salvador and develop proposals for their enhancement. The Project departs from the assumption that a common interest exists to move forward in this field and that high-level support is ensured (see output 1.1.1). The working group will take benefit from methodologies, standards and best practices worldwide and receive advice and technical assistance from international experts.[57]

83. The group will enable the stakeholders to engage and analyze urban (road) design methodologies and applied standards and parameters. Each stakeholder will assign a staff member as focal person to participate in the working group. The Project will assign a dedicated expert to VMT and OPAMSS to ensure staff capacity. A meeting schedule will be agreed under leadership of the Technical Advisor. Knowledge voids will be identified and training sessions and expert support will be provided as appropriate. It is expected that the working group will gradually take ownership of the matter, identify and prioritize key issues, and start drafting methodologies and technical standards adapted to the national context

84. Specifically, GEF funding under this output will be used for: (i) hiring of Project Technical Advisor for technical advice and supervision; (ii) hiring of two Technical Working Group Experts to act as VMT and OPAMSS focal persons; (iii) contract with national or international specialized firm/institute for updating of applicable regulation and standards.

Output 1.1.4. Capacity building for low-emission urban planning for public administration officers and other stakeholders in the central Government, OPAMSS and municipalities (GEF US$ 90,000; co-finance US$ 811,693).

85. This output will deliver training events to fill identified knowledge gaps and enhance proficiency of public officers in the use of methodologies and tools for integrated low-emission urban planning and analysis. A medium-term objective is to contribute to a paradigm shift towards a more compact urban development in the AMSS and foster multi-disciplinary, inter-institutional teams (involving MOPTVDU, OPAMSS, CNE, and municipalities), which requires building up some momentum within these institutions. Training will therefore start within a small circle of directly involved staff (focal points and working group participants), gradually expanding to colleagues and higher management.

86. Initial training activities will envisagedly be organized in-house, i.e. organized at the host's premises (VMT, OPAMSS). It is the intention to mainstream integrated urban planning, including the principles of energy efficiency, resource efficiency and low-emission mobility) into the curriculum of the national school for public administration officers (ENAFOP). This is dependent however on the evolution of ENAFOP, which has started only recently. Under leadership of CNE, opportunities will be explored (i) to draw upon existing capacities in private universities, specifically UCA and Don Bosco, (ii) to benefit from ongoing work on EE in buildings under the PESAE programme; and (iii) to integrate concepts and best practices for EE public lighting systems in municipalities.

87. GEF funding for this activity will consist of: (i) hiring of Project Technical Advisor for technical advice and supervision; (ii) hiring of two Technical Working Group Experts for VMT and OPAMSS; (iii) one or more contracts with national firm/institute for organization of promotional events and short workshops targeting public officers, policy makers, and other stakeholders.

Output 1.1.5. Design and adoption of innovative business models for mobility solutions and public lighting, including model contracts, manuals and supervision protocols (GEF US$ 80,000; co-finance US$ 200,000).

88. This output will develop or enhance business and management models for mobility and public lighting.[58] It will also explore modalities for public-private partnerships to invest in integrated urban development projects, for example commercial and leisure areas such as shopping malls where innovative mobility and public lighting solutions can add significant value for private investors.

[56] Which will invite other relevant entities within MOPTVDU such as DIIS and UGS, and the VMVDU.

[57] Including from peer institutions associated in the GEF Global Platform Sustainable Cities (GPSC).

[58] Note that the business model for SITRAMSS is targeted separately under output 1.1.2.

89. There is no experience with public bicycle systems yet in El Salvador. Some small private rental companies exist in Santa Tecla for recreational purposes. There is no data confirming sufficient demand for a public bicycle scheme and a (SWOT) analysis of organizational, technical, and operational aspects has not been made. This output will elaborate detailed proposals for business options for bicycle deployment in the AMSS based on international experiences and consultations with local stakeholders, including civil society organizations (CSOs) and the Municipality of Santa Tecla, which has specific plans into this direction. While a concession-based system is one option, another perspective is the integration of bike rental into other commercial activities along bicycle and pedestrian corridors thereby promoting SME development.[59]

90. Public lighting is a competence of the municipalities and is implemented directly or through a concessionary. Disputes between municipalities and concessionaries occur in the absence of proper performance indicators and the lack of respected arbitrage.[60] An underlying major problem is the lack of technical regulation for public lighting in El Salvador, both for equipment and for system design; this barrier is addressed specifically under output 3.1.1. In the AMSS, especially along main roads, public lighting is usually designed, implemented and financed by MOP/FOVIAL, after which property is transferred to the municipality. The AMSS municipalities typically lack know-how and financial means for adequate maintenance and appropriate retrofitting of EE lighting. Outside the AMSS, investment in public lighting is hampered by a lack of financial resources of the municipalities. Here, concession models can be used to attract private investment, as recently demonstrated by AES in Sonsonete.[61]

91. Under guidance of CNE, this output will review and enhance existing arrangements and concession models for public lighting. Within the AMSS, the interaction between municipalities and MOP is also assessed, as well as opportunities for making transfer of infrastructure more effective and efficient.[62] This output will explore more expedite arrangements, including mobilization of SME in the involved AMSS municipalities to perform maintenance services. It will draw on the Technical Working Group to systematize current weaknesses in public service contracting and work towards more expedite business models ensuring quality standards, operational and financial sustainability and transparency. GEF funding will be used for: (i) hiring of Project Finance Advisor (international consultant) for advice and development of business models; (ii) hiring of two Technical Working Group Experts for VMT and OPAMSS; and (iii) hiring of specialized firm for production of templates, guidelines and model contracts.

Output 1.1.6. Design and implementation of financing mechanism for investment in integrated low-emission urban development covering public infrastructure and buildings, mobility projects, and public-private partnerships (GEF US$ 110,000; co-finance US$ 700,000).

92. The purpose of this output is to assist the GOES and COAMSS to design and implement a more coherent and effective framework to finance a low-emission development path in the AMSS. It is noted that current finance instruments by international development banks are rather ad-hoc and insufficient to cover the large investment needs required in the AMSS. Meanwhile, targeted beneficiaries often lack the capacity to make use of credit lines because of lack of creditworthiness (SME) or regulatory constraints (municipalities). There is very little experience with partnerships between the public sector and private enterprises (PPP) to mobilize private capital for public benefits. These barriers have been identified and documented.[63]

[59] As promoted by the National Commission for Micro and Small Enterprises (CONAPYME) under the name Tienda Antena.

[60] Such a dispute exists in Santa Tecla, which makes evident that lack of administrative arbitrage causes contract problems to be easily politicized by bringing issues to court.

[61] See, for example: "Case Study of Innovative and Successful Business Models that Enable Latin American and Caribbean Cities to Adopt Efficient Technologies in Street Lighting – Executive Summary", ECONOLER, 2 October 2018. Funded by Bariloche Foundation, IDB and the GEF.

[62] It is noted that this not only affects public lighting but also artefacts to facilitate access to the public, such as pedestrian bridges, sidewalks etc. This is typically done under an ad-hoc transfer agreement between MOPTVDU and the receiving municipality. The large number of covenants and contracts are an administrative burden, economies of scale are not exploited and supervision becomes compromised, including the capacity to claim guarantees. Note that in 2014, El Salvador issued a Technical Regulation (NTS 11.69.01:14) to ensure access to public spaces, especially for persons with limited physical capabilities.

[63] See: "La Eficacia de la Participación del Sector Privado a través de la Cooperación al Desarroll en El Salvador" (draft), Global

93. This output will draw upon current initiatives by BANDESAL and international development banks (IDB, BCIE)[64] to seek opportunities to make funding windows more accessible for beneficiaries. Aspects to be considered include the capacity of intermediate agents such as financial service providers for SME and energy service companies for public and private clients, and mechanisms to provide guarantees (collateral) for SME.[65] More streamlined mechanisms are critical for upscaling and tapping into global (climate) funding. A coordinating role is foreseen for MARN and MH as part of monitoring and facilitating the implementation of the NDC. Key stakeholders are further CNE and BANDESAL, as well as the international banks active in the country.
94. GEF funding will be used for: (i) hiring of Project Finance Advisor for development of financing mechanism; (ii) hiring of two Technical Working Group Experts for VMT and OPAMSS; and (iii) one contract with national or international specialized firm/institute for the detailed design of financing mechanism(s) for investment in low-emission urban development in the AMSS.

Outcome 1.2. Information and monitoring systems for low-emission development in the AMSS have been strengthened and public awareness increased (GEF US$ 195,000; co-finance US$ 1,015,000).

95. This outcome will strengthen information and monitoring systems for low-emission urban planning in the AMSS. It will build capacities among urban planners in AMSS municipalities, OPAMSS and the central government through knowledge products and training events (output 1.2.1). It will assist COAMSS in further developing the monitoring system for urban development indicators as outlined in the AMSS Master Plan 2016 (1.2.2). Work under this outcome will draw in know-how and methodologies from the global resource base, such as available through the GEF GPSC (1.2.3). To this purpose, it is foreseen to establish a partnership with an international institution with proven expertise in the field of sustainable cities and low-emission mobility.
96. In the End-of-Project situation (EOP), it is expected that COAMSS/OPAMSS will have strengthened its capacities to monitor (low-emission) development in the AMSS as follow: (i) urban development indicators will be monitored at least bi-annually; and (ii) at least three climate change indicators will be monitored.

Output 1.2.1. Dissemination of knowledge products for low-emission urban planning targeting municipalities and project developers, and awareness raising activities (GEF US$ 80,000; co-finance US$ 500,000).

97. This output will collate and disseminate knowledge products for urban planners targeting the AMSS municipalities, OPAMSS, MOPTVDU/VMVDU and VMT, as well as private stakeholders including project developers and real estate investors. This output will draw upon international know-how such as made available by the GEF GPSC (see output 1.2.3) as well as regional experiences with Transit-Oriented-Development (TOD). It will increase awareness about the importance of policy instruments to steer urban development, such as property taxes and the issuance of land for project developers under the overall direction of a public entity. Such instruments are key for implementing the zonification of the AMSS as set forth in the AMSS Master Plan 2016. With a view on EE, this output will absorb the outcomes of Component 3 (specifically outputs 3.1.1 and 3.1.3) and bring these under the attention of stakeholders.
98. GEF funding will be used for: (i) hiring of Project TA for coordination and technical backstopping; (ii); one contract with specialized international (global) institution to enter into a Project Mobility Partnership (PMP) to assist in the design and implementation of knowledge products targeting the AMSS; and (iii) printing of manuals and brochures and publication of audiovisual products. Workshops and seminars will be important for bringing together stakeholders and decision makers; these events are expected to be organized by the Project counterparts (baseline funding).

Output 1.2.2. Fine-tuning of OPAMSS impact indicators for urban planning, (including mobility), determination of baseline status and periodic monitoring (GEF US$ 50,000; co-finance US$ 215,000).

Partnership for Effective Development Co-operation, 17 October 2018. (http://effectivecooperation.org/2018/11/el-salvador-explores-more-effective-engagement-of-the-private-sector-in-development-co-operation/).

[64] See: https://www.iadb.org/es/project/ES-L1132 (IDB) and https://www.bcie.org/modalidades-e-instrumentos-financieros/intermediacion-financiera/programas-de-mipyme/iniciativa-mipymes-verdes/ (BCIE).

[65] This responds to recommendation 3 of the Terminal Evaluation of the GEF-4 EEPB project.

99. The purpose of this output is to strengthen the framework for evaluating the social, economic and environmental development in the AMSS as presented in the 2016 AMSS Master Plan (COAMSS). Specifically, it aims to develop a set of indicators for measuring the impact of low-emission urban development policies and programmes in the AMSS. This output will build on methodologies and tools globally available and adapt these to the local context. It is envisaged to link this framework to the indicators proposed in the Master Plan[66]. Concerning mobility, the GEF VEEMT templates provide a basis for the selection of input data and verification of parameters such as: number of users per transport modality; shifts towards low-emission mobility modalities; travel time reductions; creation of new jobs for mobility services; and personnel safety indicators, among others. The design of the framework will consider the availability of data and the feasibility of mechanisms to collect these data.

100. This output will be developed in close coordination with COAMSS and MARN (possibly through CODEMET) to ensure alignment with overarching national monitoring systems and protocols, which are instrumental for progress monitoring of El Salvador's NDC. Guidance from these entities will be requested for developing specific terms of reference for contracted services. GEF funding will be used for: (i) with Project TA for coordination and technical backstopping; (ii) hiring of one M&E Expert (national consultant) for assisting OPAMSS in fine-tuning of indicators and data collections and analysis.

Output 1.2.3. Adoption by the AMSS municipalities of best practices for low-emission urban planning and MRV[67] methodologies developed by the GEF GPSC[68] (GEF US$ 65,000; co-finance US$ 300,000).

101. The GEF GPSC has developed a substantial corpse of knowledge products and toolboxes to support urban planners, covering the following topics: (i) climate change; (ii) disruptive technologies; (iii) environment; (iv) integrated urban planning; (v) transit-oriented development; (v) municipal finance and PPP; and (vii) geospatial data and indicators. This is available at the GPSC website at no cost.[69]

102. However, it must be noted that El Salvador as yet lacks the institutional framework for making effective use of these concepts and tools. The municipalities, while formally autonomous and responsible for their own (urban) planning, lack adequate human resources, financial means and spare time to familiarize with these and apply them in real situations. As such, it is deemed more convenient to introduce the GSPC products at the centralized level, i.e. through MOP and OPAMSS. Moreover, as outlined in the AMSS Master Plan, many planning issues (such as, for example, SITRAMSS, or water management) must be coordinated and concerted at the Metropolitan level. Once overall planning contours are set, the municipalities can benefit from GPSC knowledge and practices for designing local interventions. A direct application thereof will be the envisaged municipal mobility plans (see output 2.1.1). Guidance by experienced international experts with experience in the use of urban planning methodologies (such as provided by GPSC) is deemed critical to ensure the transfer of knowledge to AMSS planners and for technical backstopping to address specific situations.

103. GEF funding will be used for: (i) hiring of Project TA for coordination and technical backstopping; and (ii) hiring of specialized institution (Project Mobility Partnership) to assist in implementation of MRV methodologies and best practices from the GEF GPSC.

Component 2. Promoting energy efficiency measures for mobility in the AMSS.

Outcome 2.1. Sustainable urban mobility plans and pilots have been designed in selected AMSS municipalities (GEF US$ 625,000; co-finance US$ 4,835,000).

104. This outcome aims to prepare integrated urban mobility plans in four selected municipalities in the AMSS, clustered as follows: (1) Santa Tecla and Antiguo Cuscatlán; and: (2) San Salvador and Soyapango. It will establish a Municipal Working Group (MWG) on low-emission mobility with representatives from the municipalities, as well as from OPAMSS and MOPTVDU (output 2.1.1). It further encompasses the design of low-emission mobility

[66] See Annex L, section 6.6 for indicator list.

[67] MRV = Monitoring, Reporting and Verification. Such as considered in sectoral climate change plans and national Observatories.

[68] GPSC = Global Platform for Sustainable Cities.

[69] See: https://www.thegpsc.org/knowledge-products.

pilots as part of integrated urban development planning (2.1.2), taking benefit from know-how acquired through Component 1 (1.2.1 and 1.2.3). Gender and public safety aspects will be reviewed (output 2.1.3) and recommendations for enhancement of the pilots issued.

105. Capitalizing on the anticipated collaboration between municipalities, OPAMSS, MOPTVDU/VMT, CNE, the Project Mobility Partnership[70] and involvement of national universities, it is anticipated to establish a permanent "expertise hub" as part of the Project's exit strategy (2.1.4). Knowledge management through the collection of lessons learned will be a transversal element under this outcome. The Project strives at making information, methodologies and best practices available to AMSS municipalities while recognizing the leadership of MOPTVDU to initiate technical standards.

106. In the End-of-Project situation (EOP), the following results are expected: (i) four (4) municipal mobility plans have been developed and implemented; (ii) an expertise hub for urban mobility has been established; and (iii) partnerships have been established between national key actors (OPAMSS, MOPTVDU/VMT), international mobility institutions and national universities.

Output 2.1.1. Development and adoption of integrated, low-emission mobility plans in four municipalities in the AMSS[71] (GEF US$ 230,000; co-finance US$ 2,735,000).

107. This output will support the production of municipal mobility plans by providing technical backstopping, hands-on training and tailored studies. Besides planning and mapping of infrastructure and traffic regulation, other relevant aspects include: (i) concessions and contracting modalities; (ii) business models for implementation and operation; (iii) budget and financing plan. The Project will establish a municipal working group on low-emission mobility with representatives from four municipalities in the AMSS, as well as from OPAMSS and MOPTVDU. The partner municipalities are grouped in two as follows: (1) Santa Tecla and Antiguo Cuscatlan, characterized by a high per capita income, low population density and focalized mobility "hot spots". And: (2) San Salvador and Soyapango, with a medium to low per capita income, very high population density and generalized mobility challenges, including the SITRAMSS Phase I.

108. The purpose of the working group is to reconcile municipal-level needs and expectations with the higher-level perspectives assumed by OPAMSS and MOPTVDU, with the objective to: (i) make plans and proposed solutions as relevant as possible; (ii) understand the role of local actors (including municipality, CSOs, community groups and private sector) for delivery of solutions; (iii) collect, discuss and systematize information and parameters to establish a baseline and targets; and (iv) share and enhance approaches, methodologies and lessons across municipalities, OPAMSS and MOPTVDU. The working group provides core capacity for the development of the municipal mobility plans and act as a platform for engagement with the stakeholders groups.

109. This output will contract two national consultants to ensure adequate capacity is in place at the municipal level. The consultants will operate as a tandem under guidance of the Project Technical Advisor and take part in the municipal working group. They will set up the process of municipal mobility planning and pilot project development in the municipalities, and keep this on track in accordance with the overall planning of the GEF Project as stipulated by the Project Board. It is envisaged that each municipality will assign a focal person in charge of transport. The consultants will envisagedly be active for about 3 years each (36 months).

110. A calendar for developing the plans will be set by COAMSS/OPAMSS in consultation with the Project and UNDP. A phased approach is recommended to spread support capacity (i.e. starting in two municipalities first). The scope of the plans shall be adjusted to the local context but preferably focus on practical solutions with a short to medium term time horizon (say 2-7 years) to gain experience and demonstrate positive impact. Envisagedly, complexity will be less in Santa Tecla and Antiguo Cuscatlan than in San Salvador and Soyapango.

111. In Santa Tecla, the Juegos Centroamericanos sports event in 2021 provides an attractive entry point for a municipal mobility plan to enhance the impact of public and private investment. A trust fund has been

[70] A contractual arrangement with a renowned international sustainable mobility institution as a strategic Project partner providing policy advice, transfer of knowledge, and specific support on technical issues. As and if appropriate, the scope of the partnership can be widened to include urban planning and seeking integration into the GEF Global Platform for Sustainable Cities.

[71] The envisaged municipalities are, tentatively: Santa Tecla, Antiguo Cuscatlan, San Salvador, and Soyapango.

established to this purpose with US$ 500,000 committed by the Municipality.[72] In San Salvador and Soyapango, mobility plan development will likely involve extensive consultation with local actors. The plans shall be supportive to SITRAMSS and as such, involve traffic management solutions including preferential lanes, conditioning of SITRAMSS corridors, changes in traffic direction and road design in areas influenced by SITRAMSS (such as parallel avenues), inclusive access to SITRAMSS bus stops, and enhanced bike and pedestrian routes and services.

112. The municipal mobility plans will draw on Component 1, specifically concerning delimitation of mandates between the municipalities and MOPTVDU (output 1.1.1), the SITRAMSS Law project (1.1.2) and best practices and tools (1.2.3). Since conventional bus transport is still one of the main transport modalities, it needs to be included. This urges for the formalization and reorganization of existing bus routes under responsibility of VMT (which is overdue). A relevant baseline initiative is the preparation of SITRAMSS Corridor II (north-south) with financial inputs from MOP/FOSEP and IDB (US$ 0.54 mln).[73]

113. GEF funds will be used for: (i) hiring of Project TA for technical advice and supervision; (ii) hiring of two Municipal Working Group Experts (national consultants) to be assigned to four municipalities in the AMSS; (iii) hiring of one or more Social Engagement Experts (national consultant) to interact with stakeholder groups; and (iv) one or more contracts with national or international specialized firm/institute for the development of integrated mobility plans in four AMSS municipalities.

Output 2.1.2. Feasibility studies and technical design studies carried out for selected, innovative low-emission mobility pilots (GEF US$ 280,000; co-finance US$ 1,000,000).

114. This output will deliver prefeasibility studies, feasibility studies and technical design studies for selected low-emission mobility pilots in the AMSS. Opportunities include improvements to the SITRAMSS routes, bus stations and management system, as well as changes in traffic management in the SITRAMSS influence area. Another type of intervention are the secure pedestrian walking routes (“RAPS”) for which conceptual studies have been performed during the PPG in collaboration with MOPTVDU-DIIS, for example connecting the San Jacinto complex with central San Salvador as part of an urban rehabilitation project.[74]

115. The studies will subcontracted to specialized firms or institutions. The requested level of detail for the studies will depend on the chosen contracting modality, leaving detailed engineering and design optimization to the contractor. Smaller interventions that will be executed directly by MOPTVDU or FOVIAL will need full technical specification and design. The Project will establish a work program in dialogue with the municipalities, MOPTVDU and COAMSS. Based on recommendations issued and with consent of UNDP, a final selection of pilots will be made by the Project Board, taking into account technical merits and feasibility, impact score, level of civil society engagement, gender equity and inclusiveness, and value-added of GEF support.

116. The Project will seek to leverage the available GEF resources with other sources, including the regular MOPTVDU budget, the Salvadoran Fund for Pre-investment Studies (FOSEP), as well as multilaterals and bilaterals and other actors. It will further seek to develop jointly with private sector agents, at least one (1) integrated urban development and mobility project, tentatively around commercial (shopping and leisure) centers. Where relevant, the pilots will incorporate EE public lighting systems (separate from, or conbined with, the EE pilots pursued under 3.2.1). With a view on building a body of best practices, the Project will carefully assess the potential pilots to ensure that social, economic and environmental benefits can be actually determined and tracked.

117. GEF funds under this output will be used for: (i) hiring of Project TA for technical advice and supervision; (ii) hiring of two Municipal Working Group experts for four AMSS municipalities; and (iii) one or more contracts with national or international specialized firm/institute for the development of detailed feasibility studies and technical design studies for selected low-emission mobility pilots.

[72] See GEF CEO Endorsement Request § 52.

[73] See GEF CEO Endorsement Request § 50.

[74] See Annex M, Project 3.

Output 2.1.3. Design of proposals to address public safety issues for accessing SITRAMSS following a gender-based approach (GEF US$ 25,000; co-finance US$ 300,000).

118. It is envisaged to integrate public safety and gender-responsive measures into mobility plans (output 2.1.1) and feasibility studies for pilots (2.1.2). In this context, it is noted that El Salvador already has a technical regulation to ensure access to individuals with physical disabilities (NTS 11.69.01:14). Identified measures include: (i) improved route signaling, including an area map at each SITRAMSS bus station; (ii) adequate lighting around bus stops and along pedestrian routes (RAPS); (iii) video surveillance; (iv) proper training of security personnel at SITRAMSS bus stops and buses.

119. Concerning the latter, it is observed that the SITRAMSS is currently protected by military, who are not trained for deployment in a civilian context (with predominant female users). Additional training would be encommendable, but SITRAMSS protection needs to be regulated (see output 1.1.2) and based on a civil protection (either a SITRAMSS police or a private security service). With a view on safety, it is essential that SITRAMSS attains its design frequency soonest to avoid long passenger waits at the bus stops and reduce opportunities for delinquents to enter the stations and the buses.

120. GEF funds under this output will be used for: (i) hiring of one Safety and Gender Expert (national consultant) for design of proposals to address safety and gender concerns.

Output 2.1.4. Establishment of an expertise hub for urban mobility in El Salvador in conjunction with CNE, MOP/VMT, OPAMSS and the academic sector (GEF US$ 90,000; co-finance US$ 800,000).

121. Many cities in Latin America and around the globe have proven that public transport can be reorganized and made more comfortable, secure, effective and fuel-efficient. This output aims to increase access to state-of-the-art knowledge on sustainable mobility concepts and technologies for stakeholders in El Salvador, specifically policy makers, VMT, OPAMSS, municipalities, transport companies, educational entities, professionals, and civil society organizations. The PPG observed that few (or none) national stakeholders are affiliated to international transport associations and that in-country expertise with sustainable urban mobility development is scarce. This is a missed opportunity for accelerating the learning curve.

122. The PPG detected a lack of know-how among stakeholders to develop a long-term vision on urban mobility, evidenced by a focus on short-term solutions rather than a systemic approach. As an example, the PIF emphasizes Transit-Oriented-Development (TOD) and mass transport systems (BRT), but in parts of the AMSS such as Antiguo Cuscatlán[75], demand-driven transport solutions including (electric) taxis and small collective buses in combination with smart communication technology may prove more responsive to future needs. Moreover, demand-driven mobility solutions are scalable and tend to require less (lighter and cheaper) fixed infrastructure. They are less invasive in a dense urban space as the AMSS and can serve both people and light cargo (parceling) between shopping malls, universities, and government centers, among others. International partnerships can bring up-to-date know-how and experience to the country.

123. This output will therefore tap into the global resource base by fostering contacts between national stakeholders and their peers worldwide. Under coordination of the Technical Advisor, the Project's working groups will evaluate opportunities for collaboration with national academic institutions (such as UCA and UES) as well as the public administration school (ENAFOP).[76] International partners include for example the Institute for Transportation and Development Policy (ITDP)[77] and the International Association of Public Transport (IUTP)[78] The Project will enable national counterparts to broaden their view and promote memberships of international organizations.

124. As part of the Project's exit strategy, it is envisaged to transfer elements of academic and professional curricula from international partners to national universities and educational entities, thereby creating an in-country "hub" of expertise. The Project can bear part of the costs of such transfer, including translation and printing of

[75] Reference is made to the description of mobility situations in Antiguo Cuscatlán in Annex M, Project #2 (p.36).

[76] In this context, ENAFOP's support from the French Embassy in El Salvador is noted.

[77] See: https://www.itdp.org/

[78] UITP = Union Internationale des Transports Publics. See: http://www.uitp.org/

materials, specific training of national university teachers by their international counterparts, etc. The introduction of modern mobility concepts into academic and engineering studies shall contribute to a "paradigm shift" by introducing multidisciplinary working methodologies instead of the traditional sectorial approach. Similarly, the "hub" offers an anchor point as well for knowledge products and GPSC toolboxes covered under outputs 1.2.1 and 1.2.3.

125. GEF funds under this output will be used for: (i) hiring of Project TA for technical advice and supervision; (ii) hiring of two Municipal Working Group experts for four AMSS municipalities; and (iii) contract with Project Mobility Partnership to establish an expertise hub on urban mobility in association with national stakeholders.

Outcome 2.2. Low-emission mobility solutions have been implemented along the SITRAMSS Corridor (GEF US$ 350,000; co-finance US$ 17,930,211).

126. This project component aims to implement integrated urban mobility plans in selected municipalities in the AMSS. A number of pilots will be implemented through direct public investment by MOPTVDU, OPAMSS and other partners for which cofinance has been secured up to US$ 17.6 mln. Identified pilots under the baseline include the San Jacinto Cultural and Recreational Complex in San Salvador (US$ 7.7 mln), the Rehabilitation of the Historical Center of San Salvador (US$ 0.98 millon), and the Oriental Public Transport Terminal (US$ 8 mln).[79] It is expected that substantial additional investment can be mobilized through the Government with back-up from multilateral banks and private agents.

127. In the End-of-Project situation, it is expected that: (i) capital is leveraged for investment in integrated low-emission urban development projects along the SITRAMSS Corridor; and (ii) SITRAMSS Phase I operation has improved up to the design capacity and speed along the Corridor.

Output 2.2.1. Investment in low-emission urban mobility pilots along the BRT SITRAMSS Corridor to demonstrate their social and environmental benefits (GEF US$ 350,000; co-finance US$ 17,930,211).

128. This output encompasses investment in the low-emission mobility pilots selected and prepared under outcome 2.1. Opportunities for investment include improvements to the SITRAMSS routes, bus stations and management system, and changes in traffic management in the SITRAMSS influence area. Another type of intervention are the secure pedestrian walking routes (the so-called "RAPS"). The design of these initiatives will draw upon the municipal mobility plans as prepared under (2.1.1). The actual size and scope will depend on on the amount of capital mobilized and the implementation and supervision capacity of the stakeholders.

129. The PPG team assessed a total of 39 project ideas in the AMSS, from which 7 were elaborated further.[80] These are: (1) Urban Mobility Plan Santa Tecla center – San Benito center; (2) Interconnection of 3 traffic nodes (historical center, universities, and shopping malls) in Antiguo Cuscatlán; (3) "Soft" mobility circuit between childrens´ playground, San Salvador historical center, and San Jacinto Complex; (4) "Soft" moblity and interconnection with main traffic nodes along the 25th Avenue and Parque Cuscatlan (Antiguo Cuscatlan); (5) Mobility planning and interconnection of shopping malls, SITRAMSS terminal and the center of Soyapango; (6) Development and implementation of integrated urban development plans in Santa Tecla; and (7) Deployment of a public bicycle rental scheme under a concessionary model.

130. It is expected that substantial additional investment can be mobilized through the Government with back-up from multilateral banks, specifically the IDB which has been a long-term partner of the Government to support public transport infrastructure (including SITRAMSS). Other envisaged investments will be implemented through partnerships with NGOs (such as US-based Glasswing in the Parque Cuscatlan area in San Salvador) and with private conglomerates. In Santa Tecla, the Project seeks to create synergies with the Central American Sports Games 2021, which will attract new construction activity to the municipality.

131. GEF funds under this output will be used for: (i) investment in integrated urban mobility pilots as per specification. Where appropriate, EE public lighting will be integrated into these pilots (see output 3.2.1).

[79] See also GEF CEO Endorsement Request, § 50-51.

[80] See Project Document, Annex M.

Component 3. Enabling an energy efficient development path in AMSS municipalities

Outcome 3.1: Selected AMSS municipalities have adopted an energy-efficient development path (GEF US$ 230,000; co-finance US$ 1,300,900).

132. This outcome will establish critical conditions enabling municipalities to develop and implement energy efficiency technologies in public buildings and street lighting. These conditions are: (i) adequate technical standards and design guidelines for EE street lighting; (ii) qualified human resources for EE project development and energy management by municipalities; (iii) enforcement of technical standards, through technical assistance and quality control of municipal EE projects.

133. It will develop technical standards and design manuals for public lighting in El Salvador (3.1.1). One option is to adopt current Mexican regulation, which foresees in design guidelines, product standards including for LED lighting, and energy performance indicators differentiated per street type. (Note that standards for other appliances such as A/C and electric motors already exist in the country.)

134. Municipal building managers must be able to manage energy consumption and identify EE meassures. To this purpose, the Project will train building staff (and other energy professionals) on the design and implementation of ISO 50001 energy management systems, and will seek to certify a first batch of individuals (3.1.2). It is envisaged to work with ASIMEI as a national partner with involvement of national universities and other education entities.

135. With the aim to develop a portfolio of EE projects in municipalities, the Project will set up a task force under CNE leadership. The task force provides a platform for the municipalities, CNE and MOPTVDU to foster the exchange of information and technical know-how (3.3.3). The task force will also serve as a quality control mechanism prior to formal submission of EE project proposals for funding.

136. At End-of-Project, it is envisaged that: (i) technical standards and design manual for public lighting are developed (or adopted) and implemented; (ii) a group of public building managers and energy professionals is trained and certified to implement ISO 50001 energy management systems.

Output 3.1.1. Development of technical standards and design manuals for public lighting systems in El Salvador and sharing thereof with AMSS municipalities (GEF US$ 90,000; co-finance US$ 300,000).

137. This output will develop technical standards for public lighting systems and products as no such standards currently exist in El Salvador.[81] While some progress is made on residential lighting in the regional context, with support from the UNEP and the Mesoamerica Programme[82], this has not resulted in specific technical regulation for public lighting so far. The adoption of Mexican regulation, with includes recent regulation for LED lighting, can kick-start the process in El Salvador. The preparation and enactment of technical regulation is also included in the FIDEenergetica implementation process.[83] It is noted that both mandatory and recommended Technical Regulations (NSO and NSR) must be adhered to under public procurement by Law. However, there is no effective verification hereof at municipal level, where knowledge of EE standards is inadequate. Hence, scrutinity of public lighting projects as a condition for financing is a more effective method to ensure compliance.

138. The lack of technical regulation involves both lighting products and the design of public lighting systems. While municipalities lack any formal guidance, MOPTVDU and FOVIAL are assumed to have some internal design procedures which are applied to main road infrastructure but these are not made available to other public entities. The comprehensive manual published by the Federal Transport Secretariat (SCT)[84] can serve as a starting point for El Salvador. The manual provides performance criteria for public lighting systems in function of road type and width and addresses a large number of urban lighting situations (which must satisfy motorized traffic, pedestrian visibility and public safety criteria). The availability of a public lighting manual for use by

[81] It is noted that the PIF also envisaged A/C standards. Such standards already exist in El Salvador and are enforced by verification of equipment imports (verification of product documentation).

[82] See: https://www.unenvironment.org/news-and-stories/story/efficient-lighting-strategy-central-america.

[83] See Annex P, p. 7 and 10.

[84] Manual de Iluminación Vial – Carreteras, Boulevares, Entronques, Viaductos, Pasos a Desnivel y Túneles, Secretaría de Comunicaciones y Transportes, 2da edición (2015). México D.F.

municipalities, project developers and civil society organizations would be a major asset to ensure consistent quality of public lighting designs.

139. The recent Mexican standard NOM-013-ENER-2013[85] introduces a new performance parameter in terms of electric power per street area (DEPA)[86] and imposes a maximum value (in function of road type and road cover). Interestingly, enforcement is mandated to the energy efficiency Council (CONUEE) and not the SCT. In the context of El Salvador, this would mean the integration of EE criteria into MOPTVDU guidelines, under the purview of CNE. Whichever arrangement is chosen, it is of paramount importance that CNE and MOP agree upon a modality for effective implementation of public lighting systems in the AMSS.

140. GEF funds under this output will be used for: (i) hiring of Project TA for technical advice and supervision; (ii) hiring of Project Energy Efficiency expert (national consultant) for technical advice on energy; and (iii) short consultancies by regional experts to develop or adapt EE standards and manuals for public lighting (possibly from Mexico).

Output 3.1.2. Implementation of an ISO 50001 training and certification program for municipal building operators and energy professionals (GEF US$ 80,000; co-finance US$ 400,000).

141. Energy management is a powerful tool for defining investment in EE appliances and building retrofits and assists in modifying user behavior to attain energy reduction targets. As such, it fits into corporate strategies to control operational costs, reduce environmental externalities (including GHG footprint) and increase competitiveness. If monetary savings are controlled locally, these can be re-invested in building upgrades. Energy management strengthens local ownership which in turn, provides a basis for benchmarking of energy consumption across a building portfolio.

142. In municipalities, conditions are not in place yet to assign responsibility to building operators for their energy budgets. This will change once the FIDEnenergetica is approved. It is therefore important to create awareness of energy management systems and introduce standardized approaches. To this purpose, this output aims to create familiarity with the concepts and methodologies of ISO 50001 as a basis for developing municipal and building energy management plans, data collection and reporting protocols, and the determination of energy consumption baselines and targets. Trained municipal energy officers can identify appropriate EE measures and develop proposals for funding under FIDEnergetica.

143. This output aims to introduce the ISO50001 energy management system in the public sector in El Salvador in support of the National Energy Plan 2010-2024. It is envisaged to have one national institution (envisagedly: ASIMEI[87]) accredited under a partnership with an international entity. Building in-country capacity is deemed important to serve El Salvador's industrial and commercial EE market in the future. This output builds forth on a baseline initiative for ISO 17024 certification of professionals, pursued by CNE with the Salvadoran Accreditation Organization (OSA), ASIMEI, BANDESAL, UCA and the University Don Bosco (UDB). It is the intention to certify a first group of 30 people.

144. GEF funds will be used for: (i) hiring of Project EE expert for supervision of implementation of ISO50001 training and certification program; and (ii) contract with national or international firm/institution for implementation of ISO 50001 programme in collaboration with ASIMEI.

Output 3.1.3. Deployment of a task force by CNE, MOP, and selected AMSS municipalities to develop mature EE projects in public buildings and street lighting (GEF US$ 60,000; co-finance US$ 600,000).

145. Experience in many countries shows that development of a mature EE project portfolio is hampered by the lack of capacity in public institutions and the absence of private project developers in this sector (as other EE market, such as industry, are typically more profitable and show economies of scale). To mitigate this barrier, the Project

[85] Norma Oficial Mexicana NOM-013-ENER-2013 "Eficiencia Energética para Sistemas de Alumbrado en Vialidades", published Diario Oficial de la Federación 14 June 2013.

[86] DEPA = Densidad de potencia eléctrica para alumbrado (W/m^2).

[87] ASIMEI is the Salvadoran Association of Mechanical, Electrical and Industrial Engineers.

will establish a "task force" within CNE to support municipalities to to develop mature EE projects in public buildings and street lighting. The task force is led by a dedicated EE expert hired by the Project.

146. The work will focus on the four municipalities Santa Tecla, Antiguo Cuscatlan, San Salvador and Soyapango (other municipalities are not excluded however). A long-list of buildings and services will be compiled and prioritized in function of EE potential and other criteria defined by each municipality. Energy audits will identify potential interventions including modifications in energy contracts (contracted capacity, number of meters, change in electricity provider); energy management; replacement of equipment; passive building elements; type of service and investment contracts, including ESCOs and performance contracting.

147. It is noted that public lighting systems along the larger road arteries in the AMSS are designed and built by MOP and FOVIAL, which have the technical competences in this field. In order to design upgrades or new lighting systems, it is critical that MOP and municipalities work together to identify needs and agree upon the technical solutions. A design manual (see output 3.1.1) will be an invaluable asset for communication and for building capacity at the municipal level. CNE´s role is explained as it is responsible for energy efficiency policy and is currently the most knowledgeable public entity in this field.

148. GEF funds under this output will be used for: (i) hiring of Project EE expert for project pipeline development in AMSS municipalities, including coordination between MOP, CNE and municipal staff.

Outcome 3.2: Energy efficiency measures are being implemented by selected AMSS municipalities (GEF US$ 210,000; co-finance US$ 5,000,000).

149. This outcome encompasses the implementation of a portfolio of energy efficiency (EE) measures. Co-financing corresponding to an investment worth US$ 5 mln has been secured for EE projects by municipalities. EE investment in public buildings will be framed into an energy management approach where possible, including setting of a baseline and monitoring of performance. Public lighting will be developed under adopted technical standards and design methodologies with technical inputs from CNE, MOPTVDU, and the Project team.

150. At End-of-Project, it is envisaged that: (i) investment capital has been leveraged; and (ii) verifiable savings (MWh/yr) in electricity consumption from municipal buildings and street lighting systems are attained.

Output 3.2.1. Investment in EE measures in municipal buildings and public lighting in AMSS municipalities following enhanced public procurement guidelines (GEF US$ 210,000; co-finance US$ 5,000,000).

151. This output encompasses the implementation of identified EE measures in municipal buildings and public lighting as laid out in funding proposals (output 3.1.3). Public funding (US$ 5 mln) has been committed for investment in EE measures in public buildings and lighting by EE committees (the "COEE", now integrated into the PESAE programme). The investments will be in conformity with applicable standards and guidelines (for lighting, see output 3.1.1). Since the scope of the pilots will be defined during Project implementation, only an indicative desscription of technologies, return on investment, and expected GHG emission reductions, can be given at this stage.

152. The NAMA study (2014)[88] provides a useful reference to this purpose, as it outlines an EE investment programme in municipal buildings, as follows: (i) replacement of 20% interior and exterior lighting and retrofitting of 50%; (ii) change of refrigerants in 30% of climatization equipment; (iii) substitution of 30% of standard electric motors by EE ones; (iv) installation of thermal insulation of windows in 33% of buildings; and (v) installation of PV panels (1 kWp) in 20% of buildings. By incorporating EE investments into the scheme of an ISO 50001 energy management system additional savings will be obtained due to improved user awareness and habits and more effective maintenance. Based on these figures, a US$ 3 mln investment would yield annual energy savings of about 2,024 MWh/yr with associated GHG reductions of 1,678 tCO2e/yr.

153. The NAMA study also sets a detailed baseline of public lighting services in the country, including the AMSS, leading to the conclusion that 31% of all lights are found in just five municipalities: San Salvador (25,309 units;

[88] NAMA en el uso final de la energía del sector gubernamental – Estudio sobre los sub-sectores de alumbrado público, edificios públicos y transporte nacional, project UNDP/CNE/00075672, Alberto Galante Marcos, Luca Lo Re, e.a. Perspectives GmbH, Hamburg Germany, April 2014.

13.5% of total) and Soyapango, Santa Tecla, Santa Ana and San Miguel (32,803 units; 17.5% of total). The NAMA proposes to replace existing public lighting by the most energy-efficient technology (LED of various capacities). An anticipated investment of US$ 2 mln under the GEF project would cover 3,680 light units yielding annual electric energy savings of 1,752 MWh/yr and GHG emissions of 1,369 tCO2e/yr.[89]

154. Investments in air conditioners will be aligned with El Salvador´s obligations under the Montreal Protocol, including the Kigali Amendment (2007) to avoid the introduction of HFC-containing appliances. Although the Multilateral Fund does not directly support El Salvador, UNDP is currently disseminating the System Houses approach (developed by UNDP for the MLF), to countries in Central America including El Salvador. The projects aim at avoiding the introduction of high GWP HFCs and strive to introduce the long-term solution immediately. Under the HCFC Phase out Management Plan (HPMP) Stage 1, UNDP further provides training to SME on good refrigeration practices and alternatives to HCFCs, with a strong focus on the role of women entrepreneurs.

155. For each municipality, building and lighting service, a baseline and reporting protocol will be set up to verify the obtained energy savings, GHG reductions and monetary benefits. GEF funds under this output will be used for: (i) investment in EE pilots in AMSS municipalities as per specification. It is expected that additional investment will be leveraged during the project, presumably through FIDEnergetica and by implementing of public lighting systems under concessionary models. When appropriate, some EE public lighting projects will be combined with the mobility pilots pursued under output 2.2.1.

Component 4. Monitoring and Evaluation.

Outcome 4.1: The Project monitoring & evaluation plan has been implemented (GEF US$ 115,284; co-finance US$ 30,000).

156. Monitoring of project progress is essential for the adequate and timely delivery of results. This component covers project monitoring and oversight by UNDP in close coordination with MARN, CNE and other project partners, as well as the Mid-Term Review and Terminal Evaluation of the Project.

Output 4.1.1. A monitoring and evaluation plan is designed and implemented, including reporting on progress indicators and targets (GEF US$ 30,000; co-finance US$ 15,000).

157. This output covers the organization of an inception workshop, the definition of progress and impact indicators and the design and implementation of a detailed monitoring plan and methodology. Building forth on the initial stakeholder engagement plan (see Annex F), mechanisms for effective engagement wil be devised with a focus on "street-level" actors which are often informal, vulnerable and diffuse. An effort will be made to charter the baseline situation in more detail to identify ongoing assistance programmes (by the Government, COAMSS, agencies and NGOs) in the selected municipalities and areas. In line with this, it is anticipated to further articulate the initial gender analysis and action plan (see Annex G). It is envisaged to implement these activities during PAC review cycle and integrate the results into the Project Initiation Plan. The activities will draw on UNDP staff from CO supported by the regional bureau and one or two consultants hired by the Project. The gender plan and stakeholder engagement plan will be key instruments under the mandatory Management Plan for securing Social and Environmental Safeguards given the high risk profile of the Project that resulted from the SESP.

158. GEF funding under this output encompasses: (i) hiring of one or two consultants for finalization of the stakeholder engagement plan and the gender action plan; (ii) hiring of one international M&E specialist for guidance and support during the Project's inception phase and backstopping of the monitoring process; (iii) hiring of one or more national gender experts for annual review (before PIR); and (iv) travel costs.

Output 4.1.2. Mid-term Review and GEF Terminal Evaluation are conducted (GEF US$ 61,284; co-finance US$ 15,000).

159. This project output consists of the Mid-term Review (MTR) and the GEF terminal evaluation (TE), to be carried out by a team of independent national and international consultants. The MTR will be carried out by UNDP after 18 months of project implementation. The TE will be conducted during the last three months before operational

[89] For a more detailed description, reference is made to Annex C, p.8-11.

closure of the Project. GEF funds will be used for: (i) hiring of international, independent evaluation experts for implementing the MTR and TE; and (ii) one national consultant to participate in TE evaluation team and provide logistical support.

Output 4.1.3. Project audits are conducted (GEF US$ 24,000; co-finance US$ 0).

160. This project output encompasses annual project audits in line with UNDP guidelines. GEF funds will be used to cover the cost of these services.

Project execution and partner responsibilities

161. In order to ensure delivery, the Project will work closely with the central government entities MARN, CNE, VMT and other relevant entities within the MOPTVDU; as well as with OPAMSS. Specifically, a working group will be set up under Component 1 to review and prepare technical standards and business models. The entities will each assign a focal person to the Project for technical inputs, discussion and review. Responsibility for project implementation will be assigned to CNE which has been a successful partner for UNDP El Salvador for many years. Political responsibility will be assumed by the CNE Executive Board, which will appoint the National Project Director (envisagedly: the Minister of Environment and Natural Resources).

162. Component 2 envisions an agreement between the Government and universities to build national competences in the field of mobility, in partnership with international peer organizations and cities. Under guidance of CNE, it is proposed that MOPTVDU and MINEC assume leadership to push forward the envisaged Expertise Center, given the strategic value of mobility and logistics for national development. Negotiations can take place under the umbrella of the CNE Executive Board.

163. Under Component 3, selected municipalities, VMT and OPAMSS will work together to develop municipal mobility plans and prepare and evaluate low-emission pilots. Political backup will be secured through the Cabinet for Environmental Sustainability and Vulnerability (chaired by MARN) and the CODEMET (chaired by COAMSS). The involvement of the Cabinet ensures full country ownership of the Project's objectives to push forward the envisaged improvements to the institutional and regulatory framework for transport in the AMSS.

Expected results

Socio-economic and environmental benefits

164. The Project is committed to the delivery of social, economic and environmental benefits as a result of a portfolio of low-emission mobility and EE investments, to be implemented during its lifetime. The following table summarizes the key expected benefits. The calculation methodology is provided in Annex C. It must be noted that detailed information to calculate input parameters is not available and must be generated as part of the pre-feasibility study for each pilot project. The presented gender ratio for project beneficiaries is an educated guess as no data are available.

PROJECT SUMMARY OF BENEFITS		
1	Individuals who benefit from low-emission mobility technology	50,000 people
	Gender ratio (first estimate)	m/f = 40% / 60%
	Annual fuel savings public transport	1.36 mln liter/yr
		12,130 MWh/yr (43,666 GJ/yr)
	Annual monetary savings (fuel)	US$ 1.27 mln
	Annual GHG emission reductions	3,631 ton CO2e/yr
2	Individuals who benefit from EE measures in public buildings and public lighting	not determined – to be monitored
	Annual electricity savings	3,776 MWh/yr
	Annual monetary savings (electricity)	USD 0.64 mln

	Annual GHG emission reductions	3,047 ton CO2e/yr

165. The total direct GHG reductions are 6,678 t CO2e/yr. Over a 10-year period, accumulated benefits are of the order of 67 ktCO2e. Assuming a replication factor of 2 (one) for EE investments in public buildings and services post-project, and full operationalization of SITRAMSS Corridor I after project termination, indirect GHG emission reductions of about 195 ktCO2e are yielded. The combined GHG benefits that can be ascribed to the GEF project accrue to 262 ktCO2e. Total annual energy savings are 15.9 GWh/yr with a monetary value of US$ 1.9 mln per year. The uncertainty margin in the results is estimated at about +/- 25%.

166. The cost-effectiveness of the GEF resources (US$ 2,420,548) would be US$ 36 per ton CO2eq (direct GHG emission reductions). The cost-effectiveness based on the combined emission reductions is estimated at US$ 9 per ton CO2eq.

Stakeholder engagement plan

167. The PPG has identified the following stakeholder categories: National Government (MARN, CNE, MOPTVDU, MF); local government (COAMSS, OPAMSS, selected municipalities; COMURES). Government committees and councils (Cabinet for Environmental Sustainability and Vulnerability, CODEMET, CONASAV). The academic and educational sector (UES, UCA, UMD, ENAFOP). Sector organizations and NGOs (FECOTRANS, AEAS; CONAPYME). Local CSOs (informal merchants, women organizations, bikers associations). Private sector (Union de Empresas SITRAMSS, SUBES, and others). The Project has engaged with a number of them and will expand their involvement during Project execution.

168. An initial Stakeholder Engagement Plan is presented in Annex F. This Plan will be used as a tool for reference and further detailed during the Project's inception phase, to be updated annually. The Plan is a starting point for the design of a Project communication strategy and specific communication plans, under the responsibility of the Project Manager.

Gender equality and empowering women

169. Public transport is one of the most insecure spaces in El Salvador, especially for women. A study by FUSADES, co-funded by USAID and published in 2015, assessed the status of violence in public transport in El Salvador through the observation of high-risk routes in the AMSS and by conducting focal groups.[90] The study offers a qualitative understanding of the drivers of insecurity and violence but also stresses the lack of data to monitor the sector.

170. The general perception of the public transport system in the AMSS is described as chaotic and competitive due to causes including: (i) the concession system for assigning the bus routes to the operators; (ii) the poor condition of the buses; (iii) the chaotic situation at the bus stops; (iv) the lack of compliance with applicable laws and regulation; (v) the inadequate supervision by the authorities; (vi) the inappropriate conduct by the drivers; and (vii) the inappropriate behavior by bus users. Concrete acts of violence and aggression occur in the buses and at the bus stops, and include: (a) sexual harassment of women; (b) robbery and theft of passengers; (c) extorsion of bus operators; and (d) collusion between bus operators and criminals.

171. Overcrowding is a cause of insecurity for women who are a main group of users of public transport. At least 3.3% of all homicides registered between 2008 and 2013 happened in public transport (715 cases). Surveys among victims of violence provide additional data: 20.6% of all robberies and thefts took place in buses and another 7.5% occurred at the bus stops. Moreover, 41.1% of all delicts involving the use of weapons is committed at the bus stops in the country.

172. The PPG did not assess gender aspects of energy in public buildings and services in detail. Schools and hospitals tend to host more female than male employees. The use of elevators, kitchens, bath and washing rooms is also different according to gender and age. Public (street) lighting is particularly relevant for security in public spaces

[90] FUSADES - Prevención del Crimen en el Transporte Público en El Salvador, Junio 2015, Proyecto SolucionES, Mangai Natarajan e.a., Acuerdo de Cooperación No. AID-519-A-12-00003.

especially in residential areas and spaces with large numbers of commuters such as bus terminals, large parking lots, etc.

173. The information voids under the baseline are addressed during Project implementation by screening of project activities on gender issues, including: (i) inclusion of gender dimensions during policy review and formulation of legal instruments such as SITRAMSS Law; (ii) definition of quality standards (transport, lighting) for public services that match both male and female needs; (iii) mobilization of gender expertise available within MOPTVDU and OPAMSS, for example by inviting national experts for counselling and consultation; and (iv) ensure gender expertise is included in capacity building activities by international partners and experts.

174. A pro-gender approach includes actions such as: (v) linkages with education, social programmes, small and medium enterprise (SME) development, micro-credits; (vi) inclusion of gender dimension in the feasibility studies, supportive studies and technical designs; (vii) inclusion of gender dimensions during review and formulation of business models for public transport and public services; and (viii) ensure that the selection of pilot projects is gender-balanced.

175. A series of guidelines shall be applied to the recruitment of project staff and the organization of training and workshops: (ix) secure gender balance of the training participants; and (x) promote participation of qualified women in the training events. Overarching measures include the nomination of a gender-responsive member in the Project Board. In this context, the presence of the Consumers' Ombudsman[91] in the CNE Executive Board is worthwhile mentioning. Finally, the Project will make a continuous effort to collect data on a sex-disaggregated basis for impact evaluation.

176. An indicative Gender Analysis and indicative Gender Action Plan is attached as Annex G to this Project Document. This initial plan is to be expanded during the Project's inception phase and shall comprise a more detailed assessment of parallel (baseline) programs and activities to promote gender equality in relation to urban mobility. It shall also benefit from ongoing engagement with stakeholders and result in concrete actions. The gender plan will be one of the instruments under the mandatory Management Plan for securing Social and Environmental Safeguards given the high risk profile of the Project that resulted from the SESP. Note that some project resources have been made available under component 4 to implement the above-mentioned activities in the initial stage of the Project.

Risks and assumptions

177. The PPG team has identified eight project risks (see next table) and incorporated mitigation strategies into the Project design. Notwithstanding, the complex political and institutional context presents a series of factors that cannot be controlled by the Project, involving very significant risks to the Project (risks 1, 2 and 3). As such, the overall risk profile of the Project is also rated as "high".

178. Significant risks relate to the support from stakeholders to project outputs and their ability to collaborate (risks 4-7). These risks may translate into project outputs not or partially being realized. Underlying factors include the current poor coordination between public stakeholders and lack of proficiency with energy efficiency and low-emission (non-motorized) mobility, which are new concepts in El Salvador. These risks are, to some extent, addressed by centralizing Project implementation within one entity (CNE) and establishing working groups under close control of the Project Technical Advisor. Through these short lines, reliance on management by external partners is avoided as much as possible and the delivery of project outputs is essentially a technical process.

179. The Project assumes that AMSS municipalities have a sustained interest in the matter. Positive response during PPG was obtained in Santa Tecla and San Salvador but this may not be so everywhere. Contacts were also established with private corporations owning shopping malls in the AMSS, which demonstrated interest in improved mobility (access). Societal demand can be used for drawing attention from municipalities. The Project further assumes that central Government entities represented in the CNE Executive Board will build consensus for keeping the project on track. The Project will be aligned with the next Presidential term (2019-2024) which may prove beneficial for political continuity; however, this cannot be guaranteed.

[91] The *Defensor del Consumidor*.

180. Direct involvement of the Project in SITRAMSS implementation as suggested at PIF stage would imply a diffuse process without unclear objectives to be attained. Many SITRAMSS problems relate to systemic issues with public procurement, low institutionalization of the transport sector and vested interests which cannot be addressed by a GEF project. The PPG team has opted to avoid such approach to keep the project's processes as controllable as possible. The SITRAMSS is therefore only indirectly addressed by pursuing the SITRAMSS Law as a tangible output.

#	Description	Date Identified	Type	Impact & Probability	Countermeasures / Mgmt. response	Owner	Submitted, updated by	Last Update	Status
1	The adoption of more effective institutional arrangements in support of urban planning and urban mobility would prove not feasible.	PPG	Governance and public policy; development	P = 4 I = 4	This is a major development risk as urban planning and mobility need an effective distribution of competences between the central Government and local authorities. Changes require reflection about possible enhanced arrangements and political support to push these forward. Improvements must be approved by the Executive Organ (Presidency) and the Assembly, which is beyond direct control of the Project. This challenge is addressed by advocacy, facilitation of dialogue and senior advice under Outcome 1.1.	Project Board/ UNDP CO	UNDP CO	Submission date	Updated
2	A lack of political consensus would impede the adoption of a robust legal framework for SITRAMSS.	PPG	Governance and public policy; development	P = 4 I = 4	The scale and timeframe of SITRAMSS demands a stable development path and investment climate. Therefore, it would benefit from being moved from the political domain to the realm of public administration. An appropriate measure is to anchor SITRAMSS in national law approved by the Assembly, which would provide clarity for private investors, allow for a realistic timeframe to be set, and avoid legal setbacks as currently the case. If a SITRAMSS Law is not approved, it is unlikely that investment capital for expansion can be attracted and conventional bus routes will be reluctant to integrate into the system. The recent Presidential elections (early 2019) provide an opportunity for reducing political polarization. However, this cannot be controlled by the Project and by consequence, this risk is high. The project strategy is to keep SITRAMSS governance on the political agenda and foster a broader dialogue.	Project Board/ UNDP CO	UNDP CO	Submission date	Updated
3	Senior management of MOPTVDU, OPAMSS and municipalities would prove reluctant to engage in working groups.	PPG	Governance and public policy; implementation	P = 2 I = 5	A key element of the Project' strategy is to build and anchor capacities within key Government institutions. Since sustainable mobility is currently scattered across various entities and departments, management support is needed for allowing operational staff and experts to	Project Board/ UNDP CO	UNDP CO	Submission date	Updated

#	Description	Date Identified	Type	Impact & Probability	Countermeasures / Mgmt. response	Owner	Submitted, updated by	Last Update	Status
					collaborate in working groups. Some commitment has been given during the PPG but this needs to be put in practice, which implies a risk. Potential staffing gaps are mitigated by assigning long-term consultants to the key actors.				
4	The enactment of a trust fund for EE in municipalities would be delayed or rejected by the Assembly.	PPG	Governance and public policy; Finance	P = 2 I = 4	A Law Project for the FIDEnergetica has been prepared by CNE for presentation to the Assembly. Since the legislative process in the Assembly is beyond direct control, the Project will mitigate this risk by supporting CNE with advocacy including presentations and expert panels to increase political support. The Project will meanwhile assist in developing an EE portfolio in public buildings and public lighting, thereby increasing the demand for immediate funding (and hence increasing pressure to approve the trust fund). As a further mitigation measure – as well as an opportunity – the Project will seek broadening the finance window for sustainable urban development through engagement with MARN and MH and the appointment of a Project Finance Expert.	Project Board/ CNE	UNDP CO	Submission date	Updated
5	Project partners would fail to implement activities in accordance with UNDP requirements.	PPG	Fiduciary	P = 2 I = 3	The Project involves several key stakeholders including MARN, CNE, MOPTVDU and OPAMSS which adds to complexity. As a mitigation strategy, direct Project partners are limited to MARN and CNE who have a well-established track record with UNDP projects. Both entities bear responsibility for NIM implementation while other Government entities will be operationally involved. Strategic guidance from all partners is assured through CNE's Executive Board members. Through these measures, this fiduciary risk is deemed modest.	UNDP CO	UNDP CO	Submission date	Updated
6	Stakeholders would fail to develop a growing	PPG	Development	P = 2; I = 4	Upscaling of project benefits depends on public sector capacity to attract investment capital and	Project Manager/ Technical Advisor	UNDP CO	Submission date	Updated

#	Description	Date Identified	Type	Impact & Probability	Countermeasures / Mgmt. response	Owner	Submitted, updated by	Last Update	Status
	portfolio of high-quality EE projects.				prepare high-quality project proposals. While progress has been made since the GEF-4 EEPB project, capacities among public officers to implement energy management systems and identify investment opportunities are still limited. The Project will therefore facilitate a task force assisting municipalities to accelerate EE project development. Simultaneously, the Project will build alliances with educational entities to anchor EE training and certification of professionals in the country.				
7	The Project would fail to institutionalize capacity building due to a lack of political support.	PPG	Development	P = 2; I = 3	El Salvador is in a process of shaping its institutional landscape -both horizontally and vertically- and competences for urban planning are weak and scattered poorly coordinated. In order to mitigate this risk and avoid managing fragmented stakeholders, the Project will establish a Technical Working Group and a Municipal Working Group to make key stakeholders work together on specific initiaties (regulation, municipal planning, identification and evaluation of potential pilots). It is further envisaged to create an expertise hub hosted by an academic entity with support from the key stakeholders and with technical support from international experts. The hub should have a concentrator effect for know-how building and innovation. This approach has been followed successfully in many countries but is a novelty for El Salvador. Progress will be monitored annually by the Technical Advisor, who should inform the CNE Board to take action to ensure that momentum and technical capacity are actually building up.	Project Manager/ Technical Advisor	UNDP CO	Submission date	No change
8	Sustainability of installed mobility solutions and EE systems would be affected by the impact	PIF	Sustainability	P = 1 I = 3	There are no indications that climate change impacts would affect the viability of installed systems. Minor effects may occur due to changes in ambient parameters causing EE investments to perform slightly better or less	Project Manager/ Technical Advisor	UNDP CO	Submission date	Updated

#	Description	Date Identified	Type	Impact & Probability	Countermeasures / Mgmt. response	Owner	Submitted, updated by	Last Update	Status
	of climate change.				compared to the baseline analysis. Transport infrastructure may be affected by extreme weather events causing flooding. In line with the El Salvador's NDC, infrastructure in El Salvador should be made climate-resilient. If this principle is applied to the systems pursued under this Project, operational sustainability should be guaranteed under varying climate conditions.				

Partnerships

181. The establishment of partnerships is critical for El Salvador to pursue low-emission mobility scenarios and acquire specific technical know-how and expertise in this field, as such is currently hardly available in the country. It is envisaged to establish a partnership with an international institution specialized in sustainable mobility and public transport, and to benefit from networks of Latin American cities for peer-to-peer learning. As part of the Project´s exit strategy, the intention is to establish a mobility expertise hub with support from national universities, CNE and MOPTVDU. The hub would be instrumental for bringing together national stakeholders from Government and the academic sector enabling them to share experiences and collaborate more effectively. The Project further seeks a long-term partnership with the academic sector for capacity building and curriculum development.

182. Other envisaged partnerships involve joint public-private mobility solutions, for example for connecting shopping malls, industrial zones and educational centers in the AMSS. The Project also aims to link with NGOs and bilateral cooperation agencies targeting sustainable urban development and mobility in the AMSS. Such partnerships would involve the Municipalities, OPAMSS, private sector entities and NGOs.

183. Partnerships are tracked in the Project's Results Framework and feed into the UNDP Strategic Plan 2017-2021: "…support innovative platforms that strengthen collaboration with Governments as well as with civil society and the private sector".[92]

South-South and Triangular Cooperation (SSTrC)

The Project will build upon parallel initiatives in the field of sustainable urban mobility and energy efficiency in the region, as well as knowledge products such as developed by the GEF and UNDP, including the Global Platform for Sustainable Cities (GPSC), hosted by the World Bank.

Sustainability and scaling up

184. Given El Salvador´s weak baseline capacities and institutional framework for pushing forward sustainable urban mobility strategies, it is acknowledged that the Project may not fully achieve all of its objectives in this field. However, the approach to break down the complex development problem into smaller pieces that can be addressed (relatively) independently, is expected to enhance success and sustainability of outcomes. The adoption of low-emission mobility solutions offers a range of social, economic and environmental benefits for inhabitants in the AMSS that go beyond the economic savings and avoided GHG emissions by saving fossil fuels. The introduction of new mobility concepts however requires a visionary approach, and technical capacity to define and evaluate future scenarios that go beyond short-term solutions. The proposed mobility hub is envisaged as an asset enabling the country to develop those solutions that best respond to its future needs.

185. The Project will develop technical standards and quality criteria for mobility and public services (e.g. lighting) and enhance business and financing models making investments economically and financially more robust and sustainable. Technical sustainability is addressed by making operation and maintenance models by municipalities more robust. For EE in buildings, certification of professionals is pursued as an instrument to improve the technical quality of project proposals and reduce investment risks. The Project will advocate for an entrepreneur´s perspective at mobility solutions such as bike rental schemes.

186. Direct involvement of the Project with the SITRAMSS Bus Rapid Transit system is tailored to the preparation of a Law project for submission to the Assembly. The PPG team concluded that this will be the most effective intervention to support BRT deployment as alternatives (such as direct interaction with transport sector agents, social awareness campaigns, or investment in additional equipment and systems) would be diffuse and unlikely to yield sustainable results; moreover, engagement with these stakeholders proved cumbersome. A SITRAMSS Law, if approved, is expected to be sustainable and would be a major achievement of the Project.

[92] UNDP Strategic Plan 2017-2021, p.6

187. The upscaling potential of low-emission mobility is demonstrated by the SITRAMSS Master Plan. Corridor Phase I will transport 75,000 people; completion of this corridor would reach about 500,000 people. Other low-emission solutions, such as small buses, monorails and light trains driven by demand, can be applied in specific areas in the AMSS connecting shopping malls, industrial areas, universities and government centers. Modern solutions are often electricity-based and can be applied in industries, smaller cities and port areas outside the AMSS, and may be instrumental for implementing El Salvador's policy to become a logistical hub. Building in-country know-how in this field is essential for identifying and evaluating such options.

188. Energy efficiency by municipalities has significant upscaling potential as there are over 6,500 public buildings in the country. Also public lighting and water pumping are major energy consumers. The FIDEnergetica, proposed by CNE under the baseline, is devised to enable finance to address this market. Trained and certified EE specialists are a key factor for project portfolio development and will find a large commercial and industrial market alongside the public sector targeted directly by the Project.

V. PROJECT RESULTS FRAMEWORK

Project Title: San Salvador Low-emission Urban Development Path (PIMS 5462)					
This project will contribute to the following Sustainable Development Goal (s): Directly: SDG 7 (Affordable and Clean Energy); SDG 9 (Industry, Innovation and Infrastructure). Indirectly: SDG 1, 2 SDG 3 (Good Health and Well-Being), SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth).					
This project will contribute to the following country outcome included in the CPD 2016-2020: 2.6 Measures have been taken for reliable, sustainable and efficient energy use. Indicators: (i) Metric tons of greenhouse gas emissions avoided. (ii) Number of entities taking integral low-carbon development measures.					
This project will be linked to the following output of the UNDP Strategic Plan 2018-2021: Key signature solution 5 (Close the energy gap); IRRF 2018-2021 Indicators: Tier 1 – Impact Indicator 5 (CO2 emission per unit of value added); Tier 2 – Output Indicator: 2.5.1 (Solutions developed, financed and applied at scale for transformation to clean energy and zero-carbon development)					
	Objective and Outcome Indicators	**Baseline**	**Mid-term Target**	**End of Project Target**	**Data Collection Methods and Risks/Assumptions**
Project Objective: To promote a low-emission urban development path in the Metropolitan Area of San Salvador (AMSS).	(A) Annual direct GHG emission reductions due to (i) mobility interventions in AMSS; and (ii) EE measures in municipalities (ton CO2e/yr)	None (0)	not defined	(i) 3,631 tCO2e/yr; (ii) 3,047 tCO2e/yr	<u>Means of Verification:</u> Estimate based on project parameters and educated guess for baseline development; Official documents listing adopted policy/legal instruments; Project reports and official information (CNE, VMT, MARN, OPAMSS); Assessment of implemented projects; site visits and interviews. <u>Risks:</u> 1-8 <u>Assumptions:</u> Sustained commitment of national authorities and sector entities; project activities can be implemented as planned; EE professionals are trained and support project development;adequate technical and operational performance of installed systems.
	(B) Energy savings in (i) transport fuel (GJ/yr); and (ii) electricity (MWh/yr)	None (0)	not defined	(i) 43,666 GJ/yr; (ii) 3,776 MWh/yr	
	(C) Number of policy instruments approved to support low-emission urban development (-)	None (0)	Two (2).[93]	Three (3).[94]	
	(D) Number of people served by improved mobility and EE public buildings and services (m/f).	Baseline to be determined[95]	At least 10,000 people above baseline	At least 55,000 people above baseline.[96]	
Component/Outcome: 1.1	(1a) Status of Law projects (Legislative	No Law Project for	(i) SITRAMSS; and	(i) SITRAMSS; and (ii)	<u>Means of Verification:</u>

[93] I.e. EE trust fund FIDEnergetica and SITRAMSS Law Project

[94] Idem, plus technical standards for public lighting.

[95] Quantitative data have not been made available but considering the capacity drop due to the reduced transit speed of the BRT, daily passenger rates in the current Corridor I Phase 1 stretch are probably of the order of 20,000-25,000 persons per day (educated guess).

[96] This includes restoring SITRAMSS Phase I utilization to the projected rate of approx. 75,000 users per day (single trips). Due to the operational problems, the value as of July 2018 is estimated of the order of 20,000 passengers. Other beneficiaries include pedestrians and bicycle users, people served by improved public lighting; and users of public buildings. The numbers shall be differentiated per gender.

The policy, legal and institutional framework for integrated low-emission planning in the AMSS has been strengthened.	Decree) for (i) SITRAMSS (0/1/2) and (ii) FIDEnergetica (0/1/2)	(i) SITRAMSS (0); and (ii) FIDEnerge-tica Law Project under preparation (0)	(ii) FIDEnergetica Law Projects submitted for review (1; 1)	FIDEnergetica Law Projects approved by Assembly (2; 2).	Official documents listing adopted policy/legal instruments; interviews Project reports and official information (CNE, VMT, MARN, OPAMSS); Risks: 1, 2, 3, 7 Assumptions: Sustained commitment of national authorities and sector entities; political priorities for climate change adaptation and mitigation are maintained and strengthened; transport sector governance is gradually improving under baseline; and project activities can be implemented as planned;
	(1b) Status of financing mechanism for integrated urban development enabling a low-emission development path (0/1/2).	No financing mechanism envisaged (0)	Financing mechanisms discussed in white paper (1)	At least one financing mechanism detailed and endorsed by Government (2)	
	(1c) Number of public officers trained on low-emission urban planning (m/f)	None (0)	80 public officers (40m, 40f)	160 public officers (80m, 80f)	
Component/Outcome: 1.2 Information and monitoring systems for low-emission development in the AMSS have been strengthened and public awareness increased.	(1d) (i) Monitoring frequency of urban development indicators in the AMSS by OPAMSS (-); (ii) Number of indicators monitored (-)	Baseline as provided in the COAMSS Master Plan[97]	(No mid-term target)	(i) At least bi-annually; (ii) At least three climate change indicators included.	
Component/ Outcome 2.1 Sustainable urban mobility plans and pilots have been designed in selected AMSS municipalities	(2a) Number of municipal mobility plans developed and being implemented (-).	None (0).	Two (2) plans developed.	Four (4) plans developed and being implemented.	Means of Verification: Project reports and official information (CNE, VMT, MARN, OPAMSS); Site visits and interviews with stakeholders and beneficiaries. Risks: 1, 2, 3, 5, 7, 8 Assumptions: Sustained commitment of national authorities and sector entities; project activities can be implemented as planned; partnerships with mobility experts can be established and are effective; public and privatei nvestment is mobilized; adequate technical and operational performance of installed systems.
	(2b) Status of expertise hub for mobility in El Salvador (0/1/2)	No expertise hub (0)	Institutional set-up and partnerships defined (1)	Expertise hub established within host and supported by key stakeholders (2)	
	(2c) (i) Number of international and national partnerships established with organizations for knowledge exchange, education and professional training on urban mobility (-);	None (0)	One (1)	Three (3)	
Component/Outcome: 2.2 Low-emission mobility solutions have been implemented along the SITRAMSS Corridor.	(2d) Capital leveraged for investment in low-emission mobility in AMSS (US$)	None (USD 0)	US$ 2 mln	USS 23 mln	
	(2e) Average speed of SITRAMSS buses along Phase I corridor (km/hr)	Estimated at about 8 km/hr[98]	(No mid-term target)	Original SITRAMSS design speed 20 km/hr	

[97] See: COAMSS, Esquema Director, p.49.

[98] Precise figures for the current situation are not available. Since the BRT now shares the bus lane with conventional traffic, the average speed is likely as in the baseline situation (before SITRAMSS construction), for which a range of 5-11 km/hr is indicated in: "SITRAMSS – Mejorando el transporte público del área metropolitana de San Salvador, Nevo, M., Granada, I., Ortíz, P., IDB, 2016 (p.22)".

Component/ Outcome 3.1 Selected AMSS municipalities have adopted an energy-efficient development path.	(3a) Number of building managers and energy professionals trained and/or certified (m/f)	None (0)	60 people trained (30m, 30f)	100 people trained and 30 certified on ISO 50001 (70%m, 30%f).	Means of Verification: Project reports and official information (CNE, MARN, OPAMSS); Assessment of implemented projects; site visits and interviews. Risks: 4, 5, 6, 7, 8 Assumptions: Sustained commitment of national authorities and sector entities; project activities can be implemented as planned; EE professionals are trained and support project development; public and private investment is mobilized; adequate technical and operational performance of installed systems.
	(3b) Technical standards and design manual for public lighting developed and implemented (0/1/2).	None (0)	Technical standard and design manual proposed (1)	Technical standard and manual approved and implemented (2).	
Component/ Outcome 3.2 Energy efficiency measures are being implemented by selected AMSS municipalities.	(3c) Volume of public and private capital leveraged for investment EE measures in municipal buildings and services (US$);	None (US$ 0)	US$ 3 million;	US$ 5 million;	
	(3d) Electricity saved (MWh/yr).	None (0 MWh).	1,000 (MWh/yr)	(ii) 3,776 (MWh/yr)	
Component/ Outcome 4.1 The Project monitoring & evaluation plan has been implemented.	(4a) Follow-up on mid-term review (MTR) recommendations to enhance project effectiveness and sustainability (0/1).	No MTR (0)	MTR completed and recommenda-tions addressed (1)	MTR completed and recommendations addressed (1)	Means of Verification: MTR report; project reports. Risks: 5 Assumptions: Project activities can be implemented as planned; Project Management is aware of sustainability aspects and risks and able to define adequate mitigation measures.
	(4b) Terminal Evaluation document (0/1).	No TE (0).	No TE (0).	TE completed (1)	

VI. MONITORING AND EVALUATION (M&E) PLAN

189. The Project results as outlined in the project Results Framework (RF) will be monitored annually and evaluated periodically during project implementation to ensure the Project effectively achieves these results. Project-level Monitoring and Evaluation (M&E) will be undertaken in compliance with UNDP requirements as outlined in the UNDP POPP and UNDP Evaluation Policy. The UNDP Country Office will work with the relevant project stakeholders to ensure UNDP M&E requirements are met in a timely fashion and to high quality standards. Additional mandatory GEF-specific M&E requirements (as outlined below) will be undertaken in accordance with the GEF M&E policy and other relevant GEF policies[99].

190. In addition to these mandatory UNDP and GEF M&E requirements, other M&E activities deemed necessary to support project-level adaptive management will be agreed during the Project Inception Workshop (IW) and will be detailed in the Inception Report (IR). This will include the exact role of project target groups and other stakeholders in project M&E activities including the GEF Operational Focal Point (OFP) and national/regional institutes assigned to undertake project monitoring. The GEF OFP will strive to ensure consistency in the approach taken to the GEF-specific M&E requirements (notably the GEF Tracking Tools) across all GEF-financed projects in the country. This is preferably achieved for example by using one national institute to complete the GEF Tracking Tools for all GEF-financed projects in the country, including projects supported by other GEF Agencies.[100]

Oversight and monitoring responsibilities

191. Project Manager (PM): The Project Manager is responsible for day-to-day project management and regular monitoring of project results and risks, including social and environmental risks. The Project Manager will ensure that all project staff maintain a high level of transparency, responsibility and accountability in M&E and reporting of project results. The Project Manager will inform the Project Board, the UNDP Country Office and the UNDP-GEF RTA of any delays or difficulties as they arise during implementation so that appropriate support and corrective measures can be adopted.

192. The PM will develop Annual Work Plans (AWPs) based on the Multi-Year Work Plan (MYWP), including annual output targets to support the efficient implementation of the Project. The MYWP will be prepared during inception and annexed to the Project Document. The PM will ensure that the standard UNDP and GEF M&E requirements are fulfilled to the highest quality. This includes, but is not limited to, ensuring the results framework indicators are monitored annually in time for evidence-based reporting in the GEF PIR, and that the monitoring of risks and the various plans/strategies developed to support Project implementation (e.g. ESMP, gender action plan, stakeholder engagement plan etc.) occur on a regular basis.

193. Project Board (PB): The Project Board will take corrective action as needed to ensure the Project achieves the anticipated results. The PB will hold project reviews to assess the performance of the Project and appraise the AWP for the following year. In the Project's final year, the PB will hold an End-of-Project review to capture lessons learned and discuss opportunities for scaling-up, and to highlight results and lessons learned with relevant audiences. This final review meeting will also discuss the findings outlined in the project Terminal Evaluation (TE) report and the management response.

194. Implementing Partner (IP): The Implementing Partner (here: the National Energy Council - CNE) is responsible for providing all required information and data necessary for timely, comprehensive and evidence-based project reporting, including results and financial data, as necessary. The IP will strive to ensure project-level M&E is undertaken by national institutes, and is aligned with national systems so that the data used and generated by the Project supports national systems.

195. UNDP Country Office (CO): The UNDP Country Office will support the PM as needed, including through annual supervision missions. These missions will take place according to the schedule outlined in the AWPs. Mission reports will be circulated to the Project team and PB within one month of the mission. CO will initiate and organize key GEF M&E activities including the annual GEF PIR, the independent Mid-term Review (MTR) and the

[99] See https://www.thegef.org/gef/policies_guidelines

[100] See https://www.thegef.org/gef/gef_agencies

independent Terminal Evaluation (TE). The CO will also ensure that the standard UNDP and GEF M&E requirements are fulfilled to the highest quality.

196. The UNDP CO is responsible for complying with all UNDP project-level M&E requirements as outlined in the UNDP POPP. This includes ensuring that the UNDP Quality Assurance Assessment during implementation is undertaken annually; that annual targets at the output level are developed, and monitored and reported using UNDP corporate systems; the regular updating of the ATLAS risk log; and, the updating of the UNDP gender marker on an annual basis based on gender mainstreaming progress reported in the GEF PIR and the UNDP ROAR. Any quality concerns flagged during these M&E activities (e.g. annual GEF PIR quality assessment ratings) must be addressed by the CO and the PM.

197. The UNDP CO will retain all M&E records for this Project for up to seven (7) years after Project financial closure to support ex-post evaluations undertaken by the UNDP Independent Evaluation Office (IEO) and/or the GEF Independent Evaluation Office (GEF IEO).

198. UNDP-GEF Unit: Additional M&E and implementation quality assurance and troubleshooting support will be provided by the UNDP-GEF Regional Technical Advisor and the UNDP-GEF Directorate as needed.

Additional GEF monitoring and reporting requirements

199. Inception Workshop and Report (IW/IR): A Project Inception Workshop shall be held within two (2) months after the Project Document has been signed by all relevant parties to, amongst others with the purpose to:

a) Re-orient project stakeholders to the Project strategy and discuss any changes in the overall context that influence Project strategy and implementation;
b) Discuss the roles and responsibilities of the Project team, including reporting and communication lines and conflict resolution mechanisms;
c) Review the Results Framework and finalize the indicators, means of verification and monitoring plan;
d) Discuss reporting, M&E roles and responsibilities and finalize the M&E budget; identify national/regional institutes to be involved in project-level M&E; discuss the role of the GEF OFP in M&E;
e) Update and review responsibilities for monitoring the various project plans and strategies, including the risk log; SESP, Environmental and Social Management Plan and other safeguard requirements; project grievance mechanisms; the gender strategy; the knowledge management strategy, and other relevant strategies;
f) Review financial reporting procedures and mandatory requirements, and agree on the arrangements for the annual audit; and
g) Plan and schedule PB meetings and finalize the first year AWP.

200. The Project Manager will prepare the IR no later than one (1) month after the IW. The IR will be cleared by CO and the UNDP-GEF RTA, and will be approved by the PB.

201. GEF Project Implementation Report (PIR): The PM, CO, and the UNDP-GEF RTA will provide objective input to the annual GEF PIR covering the reporting period July (previous year) to June (current year) for each year of Project implementation. The PM will ensure that the indicators included in the project results framework are monitored annually in advance of the PIR submission deadline so that progress can be reported in the PIR. Any environmental and social risks and related management plans will be monitored regularly, and progress will be reported in the PIR.

202. The PIR submitted to the GEF will be shared with the PB. The UNDP Country Office will coordinate the input of the GEF OFP and other stakeholders to the PIR as appropriate. The quality rating of the previous year's PIR will be used to inform the preparation of the subsequent PIR.

203. Lessons learned and knowledge generation: Results from the project will be disseminated within and beyond the project intervention area through existing information sharing networks and forums. The project will identify and participate, as relevant and appropriate, in scientific, policy-based and/or any other networks, which may be of benefit to the project. The project will identify, analyse and share lessons learned that might be beneficial to the design and implementation of similar projects and disseminate these lessons widely. There will be continuous information exchange between this project and other projects, as well as regional (such as CAREC) and global platforms.

204. Independent Mid-term Review (MTR): An independent MTR process will begin approx. 18 months after Project start. The MTR findings and responses outlined in the management response will be incorporated as recommendations for enhanced implementation during the final half of the Project's duration. The terms of reference, the review process and the MTR report will depart from the standard templates and guidance prepared by the UNDP IEO for GEF-financed projects available on the UNDP Evaluation Resource Center (ERC). The GEF OFP and other stakeholders will be involved and consulted during the process. Additional quality assurance support is available from the UNDP-GEF Directorate. The final MTR report will be available in English and will be cleared by the UNDP Country Office and the UNDP-GEF Regional Technical Adviser, and approved by the Project Board.

205. Terminal Evaluation (TE): An independent terminal evaluation (TE) will take place upon completion of all major project outputs and activities. The TE process will begin three (3) months before operational closure of the Project allowing the evaluation mission to proceed while the project team is still in place, yet ensuring the Project is close enough to completion for the evaluation team to reach conclusions on key aspects such as Project sustainability. The PM will remain on contract until the TE report and management response have been finalized. The terms of reference, the evaluation process and the final TE report will follow the standard templates and guidance prepared by the UNDP IEO for GEF-financed projects available on the UNDP Evaluation Resource Center. As noted in this guidance, the evaluation will be 'independent, impartial and rigorous'. The consultants that will be hired to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. The GEF OFP and other stakeholders will be involved and consulted during the process. Additional quality assurance support is available from the UNDP-GEF Directorate. The final TE report will be cleared by the UNDP Country Office and the UNDP-GEF Regional Technical Adviser, and will be approved by the Project Board. The TE report will be publically available in English on the UNDP ERC.

206. The UNDP CO will include the planned project TE in its Evaluation Plan, and will upload the final TE report in English and the corresponding management response to the UNDP Evaluation Resource Centre (ERC). Once uploaded to the ERC, the UNDP IEO will undertake a quality assessment and validate the findings and ratings in the TE report, and rate the quality of the TE report. The UNDP IEO assessment report will be sent to the GEF IEO along with the project TE report.

207. Final Report: The Project's terminal PIR along with the terminal evaluation (TE) report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the PB during an End-of-Project review meeting to discuss lesson learned and opportunities for scaling up.

208. Other Clauses: To accord proper acknowledgement to the GEF for providing grant funding, the GEF logo will appear together with the UNDP logo on all promotional materials, other written materials like publications developed by the project, and project hardware. Any citation on publications regarding projects funded by the GEF will also accord proper acknowledgement to the GEF. Information will be disclosed in accordance with relevant policies notably the UNDP Disclosure Policy[101] and the GEF policy on public involvement[102].

[101] See http://www.undp.org/content/undp/en/home/operations/transparency/information_disclosurepolicy/

[102] See https://www.thegef.org/gef/policies_guidelines

Mandatory GEF M&E Requirements and M&E Budget

GEF M&E requirements	Primary responsibility	Indicative costs to be charged to the Project Budget[103] (US$)		Time frame
		GEF grant	Co-financing	
Output 4.1.1				
Inception Workshop	UNDP CO	5,000	2,000 (in-kind. MARN)	Within two months of project document signature
Inception Report	PM	1,000	None	Within two weeks of inception workshop
Standard UNDP monitoring and reporting requirements as outlined in the UNDP POPP	UNDP CO	None	None	Quarterly, annually
Risk management	PM, UNDP CO	0	None	Quarterly, annually
Monitoring of indicators in project results framework	PM & TA	10,000	2,000 (in-kind, MARN)	Annually before PIR
GEF Project Implementation Report (PIR)	PM, UNDP CO and UNDP-GEF team	0	6,000 (in-kind, MARN)	Annually
Lessons learned and knowledge generation	PM & TA	2,000	None	Annually
Monitoring of environmental and social risks, and corresponding management plans as relevant	PM, UNDP CO	2,000	2,000 (in-kind, MARN)	On-going
Stakeholder Engagement Plan	PM, UNDP CO	2,000	3,000 (in-kind, MARN)	On-going
Gender Action Plan	PM, UNDP CO, UNDP GEF team	8,000	0	On-going
Addressing environmental and social grievances	PM, UNDP CO	0	0	On-going
Project Board meetings	PB, UNDP CO, PM	0	0	At minimum annually
Supervision missions	UNDP CO	None[104]	None	Annually
Oversight missions	UNDP-GEF team	None[105]	None	Troubleshooting as needed
Output 4.1.2				
Mid-term GEF Tracking Tool to be updated by PM	PM & TA	0	0	Before mid-term review mission takes place.
Independent Mid-term Review (MTR) and management response	UNDP CO, PM(U) and UNDP-GEF team	20,000	5,000 (UNDP)	Between 2nd and 3rd PIR.
Terminal GEF Tracking Tool to be updated by PM	PM & TA	0	0	Before terminal evaluation mission takes place
Independent Terminal Evaluation (TE) included in UNDP evaluation plan, and management response	UNDP CO, PM(U) and UNDP-GEF team	41,284	10,000 (UNDP)	At least three months before operational closure
NIM Audit as per UNDP audit policies	UNDP CO	24,000	None	Annually
TOTAL indicative COST Excluding project team staff time, and UNDP staff and travel expenses		US$ 115,284	US$ 30,000[106]	

[103] Excluding project team staff time and UNDP staff time and travel expenses.

[104] The costs of UNDP Country Office and UNDP-GEF Unit's participation and time are charged to the GEF Agency Fee.

[105] See footnote 84.

[106] Of which 15,000 cash (UNDP co-finance) and 15,000 in-kind (MARN)

VII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

209. Implementing Partner: The Implementing Partner for this project is the National Council of Energy (CNE)

The Implementing Partner is the entity to which the UNDP Administrator has entrusted the implementation of UNDP assistance specified in this signed project document along with the assumption of full responsibility and accountability for the effective use of UNDP resources and the delivery of outputs, as set forth in this document.

210. The Implementing Partner is responsible for executing this project. Specific tasks include:

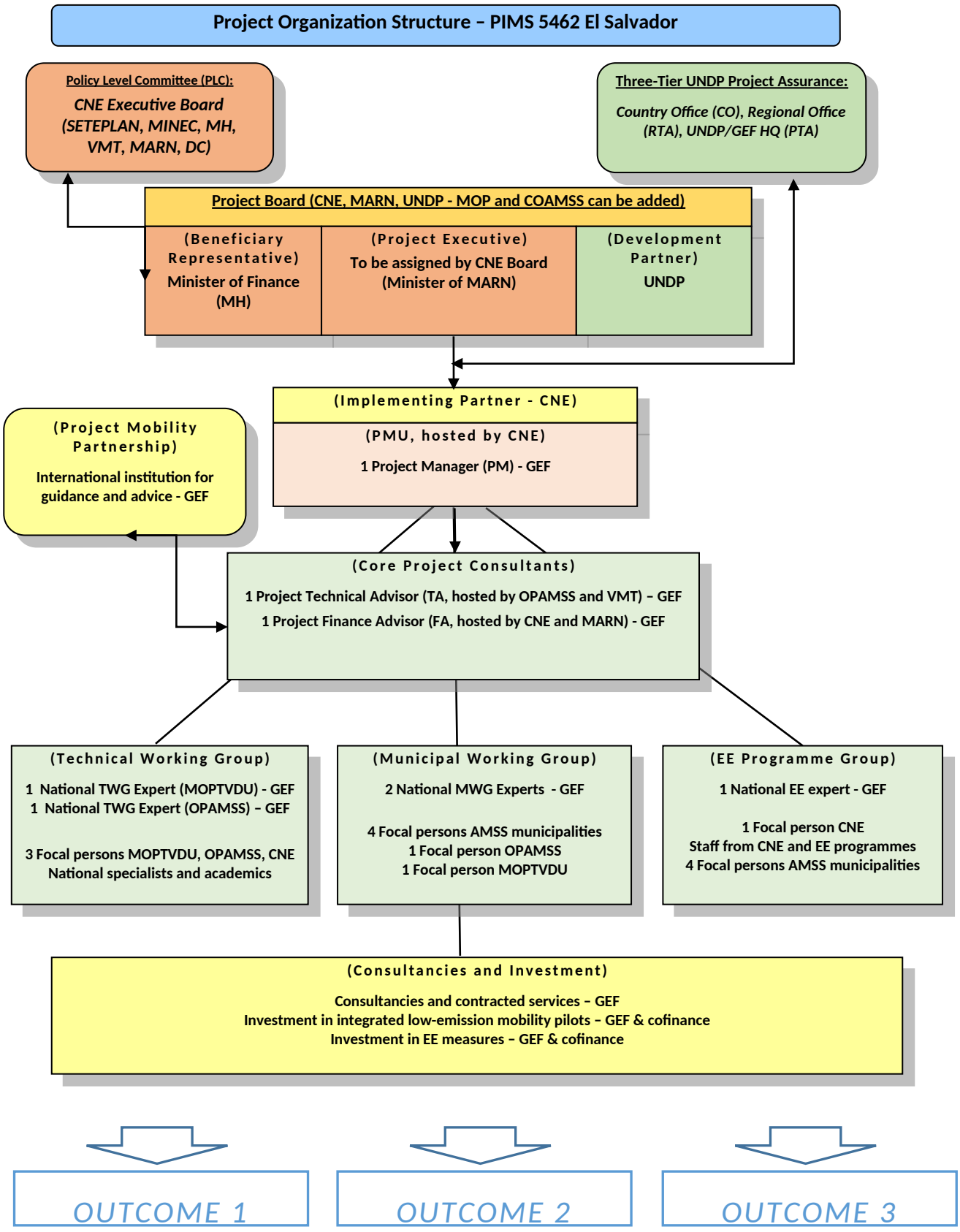
- Project planning, coordination, management, monitoring, evaluation and reporting. This includes providing all required information and data necessary for timely, comprehensive and evidence-based project reporting, including results and financial data, as necessary. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes and is aligned with national systems so that the data used and generated by the project supports national systems.
- Risk management as outlined in this Project Document;
- Procurement of goods and services, including human resources;
- Financial management, including overseeing financial expenditures against project budgets;
- Approving and signing the multiyear workplan;
- Approving and signing the combined delivery report at the end of the year; and,
- Signing the financial report or the funding authorization and certificate of expenditures.

211. Project stakeholders and target groups: The Project is targeting a variety of groups, with different levels of decision-making:

- High-level policy makers in the Executive Organ and the Assembly;
- Mayors, municipal councils and public administration staff in the AMSS;
- Transport sector staff in the MOPTVDU, specifically in its Directorates and the VMT;
- Energy sector staff in CNE and its Energy Efficiency Directorate;
- Energy professionals and mobility experts working in private sector and academic institutions;
- Entrepreneurs in the transport sector and project intervention areas in the AMSS; and
- Civil society organization including informal organizations in the project intervention areas.

212. UNDP: UNDP is accountable to the GEF for the implementation of this project. This includes oversight of project execution to ensure that the project is being carried out in accordance with agreed standards and provisions. UNDP is responsible for delivering GEF project cycle management services comprising project approval and start-up, project supervision and oversight, and project completion and evaluation. UNDP is also responsible for the Project Assurance role of the Project Board/Steering Committee.

213. Project organisation structureThe project organisation structure is as depicted in the figure:


Project Organization Structure – PIMS 5462 El Salvador
Policy Level Committee (PLC):
CNE Executive Board (SETEPLAN, MINEC, MH, VMT, MARN, DC)
Three-Tier UNDP Project Assurance:
Country Office (CO), Regional Office (RTA), UNDP/GEF HQ (PTA)
Project Board (CNE, MARN, UNDP - MOP and COAMSS can be added)
(Beneficiary Representative)
Minister of Finance (MH)
(Project Executive)
To be assigned by CNE Board (Minister of MARN)
(Development Partner)
UNDP
(Implementing Partner - CNE)
(PMU, hosted by CNE)
1 Project Manager (PM) - GEF
(Project Mobility Partnership)
International institution for guidance and advice - GEF
(Core Project Consultants)
1 Project Technical Advisor (TA, hosted by OPAMSS and VMT) – GEF
1 Project Finance Advisor (FA, hosted by CNE and MARN) - GEF
(Technical Working Group)
1 National TWG Expert (MOPTVDU) - GEF
1 National TWG Expert (OPAMSS) – GEF
3 Focal persons MOPTVDU, OPAMSS, CNE
National specialists and academics
(Municipal Working Group)
2 National MWG Experts - GEF
4 Focal persons AMSS municipalities
1 Focal person OPAMSS
1 Focal person MOPTVDU
(EE Programme Group)
1 National EE expert - GEF
1 Focal person CNE
Staff from CNE and EE programmes
4 Focal persons AMSS municipalities
(Consultancies and Investment)
Consultancies and contracted services – GEF
Investment in integrated low-emission mobility pilots – GEF & cofinance
Investment in EE measures – GEF & cofinance
OUTCOME 1
OUTCOME 2
OUTCOME 3

214. The Project Board (also called Project Steering Committee) is responsible for taking corrective action as needed to ensure the project achieves the desired results. In order to ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency and effective international competition.
215. In case consensus cannot be reached within the Board, the UNDP Resident Representative (or their designate) will mediate to find consensus and, if this cannot be found, will take the final decision to ensure project implementation is not unduly delayed.

216. Specific responsibilities of the Project Board include:

- Provide overall guidance and direction to the project, ensuring it remains within any specified constraints;
- Address project issues as raised by the project manager;
- Provide guidance on new project risks, and agree on possible mitigation and management actions to address specific risks;
- Agree on project manager's tolerances as required, within the parameters set by UNDP-GEF, and provide direction and advice for exceptional situations when the project manager's tolerances are exceeded;
- Advise on major and minor amendments to the project within the parameters set by UNDP-GEF;
- Ensure coordination between various donor and government-funded projects and programmes;
- Ensure coordination with various government agencies and their participation in project activities;
- Track and monitor co-financing for this project;
- Review the project progress, assess performance, and appraise the Annual Work Plan for the following year;
- Appraise the annual project implementation report, including the quality assessment rating report;
- Ensure commitment of human resources to support project implementation, arbitrating any issues within the project;
- Review combined delivery reports prior to certification by the implementing partner;
- Provide direction and recommendations to ensure that the agreed deliverables are produced satisfactorily according to plans;
- Address project-level grievances;
- Approve the project Inception Report, Mid-term Review and Terminal Evaluation reports and corresponding management responses;
- Review the final project report package during an end-of-project review meeting to discuss lesson learned and opportunities for scaling up.

217. The composition of the Project Board must include the following roles:

218. Project Executive: Is an individual who represents ownership of the project and chairs the Project Board. The Executive is normally the national counterpart for nationally implemented projects. The Executive for this Project will be assigned by CNE's Executive Board, which reunites the key high-level stakeholders from the Central Government. It is proposed to assign the role of the Executive (Project Director) to the Minister of Environment and Natural Resources (MARN). This choice is to be ratified by the CNE Executive Board (Junta Ejecutiva).[107]
219. The Executive is ultimately responsible for the Project, supported by the Senior Beneficiary and Senior Supplier. The Executive's role is to ensure that the Project remains focused throughout its life cycle on achieving its objectives and delivering the outputs that will contribute to higher level outcomes. The Executive has to ensure

[107] An alternative is, to assign this role to CNE's Executive Secretary.

that the project gives value for money, ensuring cost-conscious approach to the Project, balancing the demands of Senior Beneficiary and Senior Supplier.

220. Specific Responsibilities of the Executive include (as part of the above responsibilities for the PB):
- To ensure that there is a coherent project organisation structure and logical set of plans;
- To set tolerances in the AWP and other plans as required for the Project Manager;
- To monitor and control the progress of the project at a strategic level;
- To ensure that risks are being tracked and mitigated as effectively as possible;
- To brief relevant stakeholders about project progress; and
- To organise and chair Project Board meetings.

221. Beneficiary Representative: Individuals or groups representing the interests of those who will ultimately benefit from the project. Their primary function within the board is to ensure the realization of project results from the perspective of project beneficiaries. Often civil society representative(s) can fulfil this role. The Beneficiary Representative for this Project is: The Minister of Finance (MF), who can delegate this function.[108]

222. The choice of the Minister of Finance is motivated as follows: (i) energy savings in the public sector translate into direct monetary benefits for the Treasury; (ii) the Ministry of Finance is a key stakeholder for long-term sustainable development including the analysis of costs and benefits of investment in transport infrastructure; and (iii) the Ministry of Finance is a well-established institution with a clear mandate.

223. Specific Responsibilities of the Beneficiary Representative include (as part of the above responsibilities for the PB):
- To prioritize and contribute beneficiaries' opinions on Project Board decisions on whether to implement recommendations on proposed changes;
- To ensure that specification of the Beneficiary's needs is accurate, complete and unambiguous;
- To monitor Project activities at all stages to ensure that they will meet the Beneficiary's needs and that progress is made towards that target;
- To evaluate the impact of potential changes in the Project from the Beneficiary point of view; and
- To monitor risks to the beneficiaries frequently.

224. Development Partner: Individuals or groups representing the interests of the parties concerned that provide funding and/or technical expertise to the project. The Development Partner is UNDP El Salvador.

225. Specific Responsibilities of the Development Partner include (as part of the above responsibilities for the PB):
- To make sure that progress towards the outputs remains consistent from the Development Partner's perspective;
- To promote and maintain focus on the expected project output(s) from the point of view of Supplier management;
- To ensure that the Supplier resources required for the Project are made available;
- To issue Development Partner opinions in the PB on implementing recommendations on proposed changes; and
- To arbitrate on, and ensure resolution of, any Development Partner priority or resource conflicts.

226. Project Assurance: UNDP performs the quality assurance and supports the Project Board and Project Management Unit by carrying out objective and independent project oversight and monitoring functions. This role ensures appropriate project management milestones are managed and completed. The Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. UNDP provides a three –

[108] An alternative can be Minister of Economy (MINEC).

tier oversight services involving the UNDP Country Offices and UNDP at regional and headquarters levels. Project assurance is totally independent of the Project Management function.

227. Project Management: The Project will be implemented by the National Energy Council (CNE). It is proposed that the role of the National Project Director (NPD) is assumed by the Minister of Environment and Natural Resources (MARN); the NPD holds formal ownership of the Project. This choice is to be ratified by CNE's Executive Board, of which MARN is one of the members.

228. A dedicated Project Management Unit (PMU) will be established. The PMU will consist of a full-time Project Manager (PM) funded by the Project, and a Procurement Officer (PO) funded by UNDP, who can be full- or part-time. The PMU will be supported by a full-time Technical Advisor (TA) dedicated to mobility. Within CNE, additional capacity is created by the appointed of an EE Expert (EE) funded by the Project. The PMU will be hosted by CNE which will provide office space and make available telecommunication and internet facilities; the Project can fund essential office tools (laptop, digital camera) if needed. Travel arrangements will be made as needed in accordance with UNDP and Government guidelines for safety and efficiency. The TA will hold an office at VMT and OPAMSS enabling him/her to interact closely with the key transport stakeholders and have the necessary presence for effectively leading the working groups.

229. CNE is chosen as the PMU host because: (i) it is mandated for electricity and fuel efficiency; (ii) its internal organization is fairly straightforward with short communication lines; (iii) as a central Government body it has an operational budget to assume co-financing commitments; and (iv) CNE and UNDP have long-term working relations, which is an asset for efficient project execution.[109] As such, it is expected that the tandem PM/TA will cover urban mobility and facilitate interaction between the key stakeholders; another tandem is formed by the PM and the EE Trust Fund expert to push forward the energy efficiency agenda.

230. Among other duties, the Technical Advisor will facilitate and structure the engagement with high-level stakeholders and provide orientation on sustainable urban mobility and facilitate liaison on behalf of UNDP and the Project. The TA will be responsible for implementing the project components 1 and 2. Specifically, the TA will orient and supervise contracted services. The TA will be team leader and resource person for the Technical Working Group experts (TWE) that will be assigned to MOPTVDU and OPAMSS (2 persons) and the Municipal Working Group experts (MWE) assigned to selected municipalities (2 persons).

231. Project extensions: The UNDP-GEF Executive Coordinator must approve all project extension requests. Note that all extensions incur costs and the GEF project budget cannot be increased. A single extension may be granted on an exceptional basis and only if the following conditions are met: one extension only for a project for a maximum of six months; the project management costs during the extension period must remain within the originally approved amount, and any increase in PMC costs will be covered by non-GEF resources; the UNDP Country Office oversight costs during the extension period must be covered by non-GEF resoruces.

VIII. FINANCIAL PLANNING AND MANAGEMENT

232. The total cost of the project is USD 37,022,452. This is financed through a GEF grant of USD2,420,548, and USD 34,601,904 in confirmed co-financing. UNDP, as the GEF Implementing Agency, is responsible for the oversight of the GEF resources and the cash co-financing transferred to UNDP bank account only.

[109] As an alternative the PMU can be hosted by MARN. However, this choice would lead to fragmentation of the Project without additional benefits.

233. Confirmed co-financing: The actual realisation of project co-financing will be monitored during the Mid-Term Review and Terminal Evaluation process and reported to the GEF. The planned parallel co-financing will be as follows:

SOURCES OF PROJECT PARALLEL CO-FINANCING		
Name of co-financier	**Type of co-financing**	**Amount (USD)**
1. Ministry of Environment and Natural Resources (MARN)	in-kind	500,000
2. National Energy Council (CNE)	investment	5,000,000
	in-kind	7,000,000
3. Ministry of Public Works, Transport, Housing and Urban Development (MOPTVDU)	investment	16,630,211
	investment	541,693
	in-kind	500,000
4. Planning Office of the AMSS (OPAMSS)	investment	1,300,000
	investment	2,500,000
5. Municipality of Santa Tecla	in-kind	500,000
6. United Nations Development Programme (UNDP)	grants	100,000
	in-kind	30,000
	TOTAL:	**34,601,904**

234. Implementing Partner (IP) request for UNDP to provide country support services. The Implementing Partner and GEF OFP have requested UNDP to provide support services in the amount of *USD$ 25,000* for the full duration of the project, and the GEF has agreed to this request. The **request letter** (signed by the GEF OFP and the IP) detailing these support services are included in Annex. To ensure the strict independence required by the GEF and in accordance with the UNDP Internal Control Framework, these execution services will be delivered independent from the GEF-specific oversight and quality assurance services (i.e. not done by same person to avoid conflict of interest).

235. Budget Revision and Tolerance: As per UNDP requirements outlined in the UNDP POPP, the project board will agree on a budget tolerance level for each plan under the overall annual work plan allowing the project manager to expend up to the tolerance level beyond the approved project budget amount for the year without requiring a revision from the Project Board.

236. Should the following deviations occur, the Project Manager/CTA and UNDP Country Office will seek the approval of the BPPS/GEF team to ensure accurate reporting to the GEF:

a. Budget re-allocations among components in the project budget with amounts involving 10% of the total project grant or more;

b. Introduction of new budget items that exceed 5% of original GEF allocation.

237. Any over expenditure incurred beyond the available GEF grant amount will be absorbed by non-GEF resources (e.g. UNDP TRAC or cash co-financing).

238. Audit: The project will be audited as per UNDP Financial Regulations and Rules and applicable audit policies. Audit cycle and process must be discussed during the Inception workshop.

239. Project Closure: Project closure will be conducted as per UNDP requirements outlined in the UNDP POPP. All costs incurred to close the project must be included in the project closure budget and reported as final project commitments presented to the Project Board during the final project review. The only costs a project may incur following the final project review are those included in the project closure budget.

240. Operational completion: The project will be operationally completed when the last UNDP-financed inputs have been provided and the related activities have been completed. This includes the final clearance of the Terminal Evaluation Report (that will be available in English) and the corresponding management response, and the end-

of-project review Project Board meeting. **Operational closure must happen with 3 months of posting the TE report to the UNDP ERC**. The Implementing Partner through a Project Board decision will notify the UNDP Country Office when operational closure has been completed. At this time, the relevant parties will have already agreed and confirmed in writing on the arrangements for the disposal of any equipment that is still the property of UNDP.

241. Transfer or disposal of assets: In consultation with the Implementing Partner and other parties of the project, UNDP is responsible for deciding on the transfer or other disposal of assets. Transfer or disposal of assets is recommended to be reviewed and endorsed by the project board following UNDP rules and regulations. Assets may be transferred to the government for project activities managed by a national institution at any time during the life of a project. In all cases of transfer, a transfer document must be prepared and kept on file[110]. The transfer should be done before Project Management Unit complete their assignments.

242. Financial completion (closure): The project will be financially closed when the following conditions have been met: a) the project is operationally completed or has been cancelled; b) the Implementing Partner has reported all financial transactions to UNDP; c) UNDP has closed the accounts for the project; d) UNDP and the Implementing Partner have certified a final Combined Delivery Report (which serves as final budget revision).

243. The project will be financially completed **within 6 months of operational closure or after the date of cancellation**. Between operational and financial closure, the implementing partner will identify and settle all financial obligations and prepare a final expenditure report. The UNDP Country Office will send the final signed closure documents including confirmation of final cumulative expenditure and unspent balance to the BPPS/GEF Unit for confirmation before the project will be financially closed in Atlas by the UNDP Country Office.

244. Refund to GEF: Should a refund of unspent funds to the GEF be necessary, this will be managed directly by the BPPS/GEF Directorate in New York. No action is required by the UNDP Country Office on the actual refund from UNDP project to the GEF Trustee.

[110] See https://popp.undp.org/_layouts/15/WopiFrame.aspx?sourcedoc=/UNDP_POPP_DOCUMENT_LIBRARY/Public/PPM_Project%20Management_Closing.docx&action=default.

IX. TOTAL BUDGET AND WORK PLAN

Total Budget and Work Plan			
Atlas Award ID:	00107731	Atlas Output Project ID:	00107946
Atlas Proposal or Award Title:	Desarrollo Urbano Sostenible en San Salvador		
Atlas Business Unit	SLV0		
Atlas Primary Output Project Title	Desarrollo Urbano Sostenible e		
UNDP-GEF PIMS No.	5462		
Implementing Partner	National Council of Energy		

Atlas Activity (GEF Component)	**Atlas Implementing Agent**	**Atlas Fund ID**	**Donor Name**	**Atlas Budgetary Account Code**	**ATLAS Budget Account Description**	**Amount Year 1 (USD)**	**Amount Year 2 (USD)**	**Amount Year 3 (USD)**	**Amount Year 4 (USD)**	**Amount Year 5 (USD)**	**Total (USD)**	**See Budget Note:**
COMPONENT 1 Enabling framework for low-emission urban development.	**CNE**	**62000**	**GEF Trustee**	71200	International Consultants	52,000	70,000	70,000	40,000	0	**232,000**	1
				71300	Local Consultants	29,000	40,000	30,000	30,000	10,000	**139,000**	2
				71600	Travel	10,000	11,000	10,000	8,000	3,000	**42,000**	3
				72100	Contractual Services – Companies	49,000	140,000	100,000	30,000	0	**319,000**	4
				72200	Equipment and Furniture	0	0	0	0	0	**0**	**5**
				72800	Information Technology Equipment	0	0	0	0	0	**0**	6
				74200	Audio Visual & Print Prod Cost	10,000	10,000	5,000	1,000	1,000	**27,000**	7
				74500	Miscellaneous Expenses	5,000	5,000	4,000	2,000	0	**16,000**	8
					Total Outcome 1	**155,000**	**276,000**	**219,000**	**111,000**	**14,000**	**775,000**	
COMPONENT 2: Promoting energy efficiency measures for mobility in the AMSS	**CNE**	**62000**	**GEF Trustee**	71200	International Consultants	9,000	18,000	18,000	8,000	0	**53,000**	9
				71300	Local Consultants	28,000	40,000	40,000	40,000	10,000	**158,000**	10
				71600	Travel	5,000	5,000	5,000	5,000	0	**20,000**	11
				72100	Contractual Services – Companies	0	120,000	120,000	100,000	40,000	**380,000**	12

				72200	Equipment and Furniture	0	80,000	135,000	109,000	30,000	**354,000**	13
				72800	Information Technology Equipment	0	0	0	0	0	**0**	14
				74200	Audio Visual & Print Prod Cost	2,000	2,000	2,000	1,000	0	**7,000**	15
				74500	Miscellaneous Expenses	1,000	1,000	1,000	0	0	**3,000**	16
					Total Outcome 2	**45,000**	**266,000**	**321,000**	**263,000**	**80,000**	**975,000**	
COMPONENT 3: **Enabling an energy efficient development path in AMSS municipalities**	**CNE**	**62000**	**GEF Trustee**	71200	International Consultants	10,000	10,000	10,000	3,000	0	**33,000**	17
				71300	Local Consultants	16,000	25,000	25,000	25,000	10,000	**101,000**	18
				71600	Travel	7,000	5,000	5,000	0	0	**17,000**	19
				72100	Contractual Services – Companies	5,000	30,000	30,000	0	0	**65,000**	20
				72200	Equipment and Furniture	0	50,000	50,000	65,000	50,000	**215,000**	21
				72800	Information Technology Equipment	0	0	0	0	0	**0**	22
				74200	Audio Visual & Print Prod Cost	1,000	1,000	0	0	0	**2,000**	23
				74500	Miscellaneous Expenses	2,000	2,000	1,000	1,000	1,000	**7,000**	24
					Total Outcome 3	**41,000**	**123,000**	**121,000**	**94,000**	**61,000**	**440,000**	
COMPONENT 4: **Monitoring and Evaluation**	**CNE**	**62000**	**GEF Trustee**	71200	International Consultants	13,284	0	15,000	0	33,000	**61,284**	25
				71300	Local Consultants	8,000	0	5,000	0	5,000	**18,000**	26
				71600	Travel	4,000	0	3,000	0	5,000	**12,000**	27
				74100	Professional Services	4,000	5,000	5,000	5,000	5,000	**24,000**	28
					Total Outcome 4	**29,284**	**5,000**	**28,000**	**5,000**	**48,000**	**115,284**	
				71200	Local Consultants	16,264	18,000	18,000	17,000	16,000	**85,264**	29
				71600	Travel	1,000	0	1,000	0	0	**2,000**	30

PROJECT MANAGEMENT UNIT	CNE	62000	GEF Trustee	72200	Equipment and Furniture	2,000	0	0	0	0	**2,000**	31
				74500	Miscellaneous Expenses	1,000	0	0	0	0	**1,000**	32
				74596	Services to projects - GOE for CO	5,000	5,000	6,000	6,000	3,000	**25,000**	33
					Total Management	**25,264**	**23,000**	**25,000**	**23,000**	**19,000**	**115,264**	
					PROJECT TOTAL	**295,548**	**693,000**	**714,000**	**496,000**	**222,000**	**2,420,548**	

Summary of Funds:

	Amount Year 1	Amount Year 2	Amount Year 3	Amount Year 4	Amount Year 5	Total
GEF	295,548	693,000	714,000	496,000	222,000	2,420,548
parallel co-financing	1,970,211	2,500,000	10,308,000	13,300,000	6,523,693	34,601,904
TOTAL	**2,265,759**	**3,193,000**	**11,022,000**	**13,796,000**	**6,745,693**	**37,022,452**

Budget note	Comments (stated budgets are in US$)
COMPONENT 1 (GEF US$ 775,000)	
1	(1.1.1;60k$) Contract for hiring Project Technical Advisor (TA) (international consultant) to provide technical advice and facilitate the dialogue process according to TOR; (1.1.2;25k$) Contract with Project TA for advocacy and liaison; (1.1.3;17k$) Contract with Project TA for technical advice and supervision; (1.1.4;16k$) Contract with Project TA for technical advice and supervision. (1.1.5;40k$) Contract with Project Finance Advisor (FA) (international consultant) for advice and development of business models; (1.1.6;30k$) Contract with Project FA for development of financing mechanism; (1.2.1;19k$) Contract with Project TA for coordination and technical backstopping according to TOR; (1.2.2;10k$) Contract with Project TA for coordination and technical backstopping according to TOR; (1.2.3;15k$) Contract with Project TA for coordination and technical backstopping according to TOR.
2	(1.1.1;28k$) Two contracts for hiring Technical Working Group Experts (TWE) (national consultants) to be assigned to VMT and OPAMSS as focal persons, as per TOR; (1.1.2;20k$) Two TWE for VMT and OPAMSS as focal persons, as per TOR; (1.1.3;15k$) Two TWE for VMT and OPAMSS as focal persons, as per TOR; (1.1.4;18k$) Two TWE for VMT and OPAMSS as focal persons, as per TOR; (1.1.5;10k$) Two TWE for VMT and OPAMSS as focal persons, as per TOR; (1.1.6;10k$) Two TWE for VMT and OPAMSS as focal persons, as per TOR. (1.2.2;18k$) Two TWE for VMT and OPAMSS as focal persons, as per TOR; (1.2.2;20k$) Contract with one M&E Expert (ME) (national consultant) for assisting OPAMSS in fine-tuning of indicators and data collections and analysis.
3	(42k$) Mission costs (international travel and DSA) for international consultants. Costs of domestic travel (air tickets, land travel, fuel, DSA).
4	(1.1.1;30k$) One or more contracts with specialized firms or individuals to provide inputs and advice on sector governance; (1.1.2;30k$) One specialized (legal) firm for legal advice and finalization of Law Proposals for SITRAMSS and FIDEnergetica;

	(1.1.3;50k$) One contract with national or international specialized firm/institute for updating of applicable regulation regulation and standards; (1.1.4;40k$) One or more contracts with national firm/institute for organization of promotional events and short workshops targeting public officers, policy makers, and other stakeholders; (1.1.5;23k$) One contracted service for production of templates, guidelines and model contracts. (1.1.6;56k$) One contract with national or international specialized firm/institute for the detailed design of financing mechanism(s) for investment in low-emission urban development in the AMSS; (1.2.1;50k$) One contract with specialized international (global) institution to enter into a Project Mobility Partnership (PMP) to assist in the design and implementation of knowledge products targeting the AMSS, as per TOR. (1.2.3;40k$) Contract with Project Mobility Partnership (PMP) to assist in implementation of MRV methodologies, capacity building and transfer of best practices from the GEF GPSC.
5	None
6	None
7	(27k$) Printing and publications
8	(16k$) Miscellaneous expenses
COMPONENT 2 (GEF US$ 975,000)	
9	(2.1.1; 23k$) Contract with Project TA for technical advice and supervision; (2.1.2; 20k$) Contract with Project TA for technical advice and supervision; (2.1.4; 10k$) Contract with Project TA for technical advice and supervision.
10	(2.1.1;80k$) Two contracts for hiring Municipal Working Group Experts (MWE) (national consultants) to be assigned as focal persons to four municipalities in the AMSS, as per TOR; (2.1.2;20k$) Two MWE for four AMSS municipalities, as per TOR; (2.1.4;10k$) Two MWE for four AMSS municipalities, as per TOR. (2.1.1;25k$) Contract with one or more Social Engagement Experts (SE) (national consultant) to interact with stakeholder groups, as per TOR. (2.1.3;23k$) Contract with one Safety and Gender Expert (SG) (national consultant) for design of proposals to address safety and gender concerns.
11	(20k$) Mission costs (international travel and DSA) for international consultants. Costs of domestic travel (air tickets, land travel, fuel, DSA)
12	(2.1.1;90k$) One or more contracts with national or international specialized firm/institute for the development of integrated mobility plans in four AMSS municipalities, as per TOR; (2.1.2;200k$) One or more contracts with national or international specialized firm/institute for the development of detailed feasibility studies and technical design studies for selected low-emission mobility pilots, as per TOR; (2.1.4;90k$) Contract with Project Mobility Partnership (PMP) to establish an expertise hub on urban mobility in association with national stakeholders, as per TOR.
13	(2.1.1; 4k$) Office equipment (laptops and digital cameras); (2.2.1; 350k$) Investment in urban mobility pilots as per TOR.
14	None
15	(7k$) AV costs and promotional material.
16	(3k$) Miscellaneous expenses.
COMPONENT 3 (GEF US$ 440,000)	
17	(3.1.1; 13k$) Contract with Project TA for technical advice and supervision; (3.1.1; 20k$) Short consultancies by regional experts to develop or adapt EE standards and manuals for public lighting.
18	(3.1.1; 35k$) Contract with one Project Energy Efficiency expert (EE) (national consultant) for technical advice on energy S&L enforcement, as per TOR; (3.1.2; 11k$) Contract with Project EE for supervision of implementation of ISO50001 training and certification program, as per TOR; (3.1.3; 55k$) Contract with Project EE for energy efficiency project pipeline development in AMSS municipalities, as per TOR.
19	(17k$) Mission costs (international travel and DSA) for international consultants. Costs of domestic travel (air tickets, land travel, fuel, DSA)

20	(3.1.2;65k$) One contract with with national or international firm/institution for implementation of ISO50001 programme in collaboration with ASIMEI, as per TOR.
21	(3.1.1;5k$) Office equipment; (3.2.1;210k$) Investment in EE pilots in AMSS municipalities as per TOR.
22	None
23	(2k$) Print costs
24	(7k$) Miscellaneous expenses
OUTCOME 4 (GEF US$ 115,284)	
25	(4.1.1; 15k$) Contract with Project TA for assisting in project inception phase and setting up M&E Plan. (4.1.2;46,284$) Two independent Evaluation Experts (EV) (international consultants), (i) for MTR including reporting; and (ii) to lead TE team and conduct evaluation, including reporting, according to TOR.
26	(4.1.1;10k$) One M&E Expert (ME) (national consultant) for assisting in project inception phase and setting up M&E Plan.. (4.1.2; 8k$) One M&E Expert (ME) (national consultant) to participate in TE evaluation team and provide logistical support.
27	(12k$) Mission costs (international travel and DSA) for international consultants. Costs of domestic travel (air tickets, land travel, fuel, DSA).
28	(4.1.3;24k$) One or more service contracts for annual project audits as per indicated in the UNDP financial rules and regulations.
PROJECT MANAGEMENT (GEF US$ 115,264)	
29	(85,264$) One Project Manager (PM) (national consultant, full-time) as per TOR.
30	(2k$) Costs of domestic travel (air tickets, land travel, fuel, DSA) for visiting project sites and regular monitoring of contracted project activities.
31	(2k$) Two laptops, printer and digital camera.
32	(1k$) Miscellaneous expenses
33	(25k$) Direct project costs for UNDP project services as defined in the Letter of Agreement (LOA), Annex K

IDENTIFIED RESOURCES FOR PROJECT EL SALVADOR SUD (PIMS 5462)						
RESOURCE			NAME/DESCRIPTION	FUNDING SOURCE	BUDGET (US$)	BUDGET LINE
COFINANCE HUMAN RESOURCES AND OFFICE FACILITIES						
1	Project National Director	NPD	Minister of MARN	cofinance in-kind		
2	Project Board	PB	PB Member (CNE, MARN, UNDP)[111]	cofinance in-kind		
3	Programme Officer	PO	UNDP CO Programme Officer	UNDP CO (agency fee)		
4	Project Assistant	PA	UNDP CO support staff[112]	UNDP CO		
5	Procurement Officer	PO	UNDP CO support staff	UNDP CO		
6	Partner focal person CNE	CNE	CNE human resources	cofinance in-kind		
7	Partner focal person MOPTVDU/VMT	MOP	MOPTVDU/VMT human resources	cofinance in-kind		
8	Partner focal person OPAMSS	OPM	CNE human resources	cofinance in cash		

[111] MOPTVDU and COAMSS can be added.

[112] This because GEF funds for project management are insufficient for hiring a (part-time) Project Assistant and Financial Assistant.

Identified Resources for Project El Salvador SUD (PIMS 5462)						
Resource			Name/Description	Funding Source	Budget (US$)	Budget Line
9	Project Office Facilities	CNEO	CNE office and logistical support	cofinance in-kind		
10	Partner focal person municipalities (4)	MUN	Municipalities human resources (4)	cofinance in-kind		
GEF Project Resources (Long-Term Consultancies and Partnerships)						
11	Project Manager	PM	To be recruited	GEF budget	US$ 85,264	71300
12	Project Technical Advisor	TA	To be recruited	GEF budget	US$ 253,000	71200
13	Project Finance Advisor	FA	To be recruited	GEF budget	US$ 70,000	71200
14	Project Technical Working Group Expert (2)	TWE	To be recruited (for OPAMSS and MOPTVDU/VMT)	GEF budget	US$ 119,000	71300
15	Project Municipal Working Group Expert (2)	MWE	To be recruited for municipalities (Santa Tecla, Antiguo Cuscatlan, San Salvador, Soyapango)	GEF budget	US$ 110,000	71300
16	Project EE expert	EE	To be recruited for CNE task force	GEF budget	US$ 101,000	71300
17	Project Mobility Partnership	PMP	To be procured (contracted service)	GEF budget	US$ 160,000	72100
				Subtotal	**US$ 908,264**	
GEF Project Resources (Short-Term Consultancies, Services and Equipment)						
18	Social Engagement Experts	SE	To be recruited	GEF budget	US$ 25,000	71300
19	M&E Expert	ME	To be recruited	GEF budget	US$ 38,000	71300
20	Safety and Gender Expert	SG	To be recruited	GEF budget	US$ 23,000	71300
21	Technical Standards Expert	TS	To be recruited	GEF budget	US$ 20,000	71200
22	Evaluation Expert (2)	EV	To be recruited	GEF budget	US$ 46,284	71200
23	Specialized Services Companies	CSS	To be procured (contracted service)	GEF budget	US$ 604,000	72100
24	Pilot Projects Investment	PP	To be procured (equipment)	GEF budget	US$ 560,000	72200
				Subtotal	**US$ 1,316,284**	
GEF Project Resources (Other Expenditures)						
25	Travel			GEF budget	US$ 93,000	71600
26	Equipment and Furniture			GEF budget	US$ 11,000	72200
27	Audio Visual & Printing			GEF budget	US$ 36,000	74200
28	Professional Services – Audits			GEF budget	US$ 24,000	74100
29	Project – Direct Costs			GEF budget	US$ 25,000	74599
30	Miscellaneous			GEF budget	US$ 27,000	74500
				Subtotal	**US$ 216,000**	

IDENTIFIED RESOURCES FOR PROJECT EL SALVADOR SUD (PIMS 5462)				
RESOURCE	NAME/DESCRIPTION	FUNDING SOURCE	BUDGET (US$)	BUDGET LINE
		TOTAL (GEF)	**US$ 2,420,548**	

X. LEGAL CONTEXT

245. This Project Document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement (SBAA) between the Government of El Salvador and UNDP, signed on March 21, 1975[113]. All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner." The project "San Salvador Low-emission Urban Development Path" will be implemented by the National Energy Council (CNE) ("Implementing Partner") in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

246. The designations employed and the presentation of material on this map do not imply the expression of any opinion whatsoever on the part of the Secretariat of the United Nations or UNDP concerning the legal status of any country, territory, city or area or its authorities, or concerning the delimitation of its frontiers or boundaries.

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XI. RISK MANAGEMENT

247. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:

a) put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;

b) assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

248. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document.

249. The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via http://www.un.org/sc/committees/1267/aq_sanctions_list.shtml.

250. The Implementing Partner acknowledges and agrees that UNDP will not tolerate sexual harassment and sexual exploitation and abuse of anyone by the Implementing Partner, and each of its responsible parties, their respective sub-recipients and other entities involved in Project implementation, either as contractors or subcontractors and their personnel, and any individuals performing services for them under the Project Document.

(a) In the implementation of the activities under this Project Document, the Implementing Partner, and each of its sub-parties referred to above, shall comply with the standards of conduct set forth in the Secretary General's Bulletin ST/SGB/2003/13 of 9 October 2003, concerning "Special measures for protection from sexual exploitation and sexual abuse" ("SEA").

[113] Ratified by the Legislative Assembly on May 16, 1975 (Decree No. 261).

(b) Moreover, and without limitation to the application of other regulations, rules, policies and procedures bearing upon the performance of the activities under this Project Document, in the implementation of activities, the Implementing Partner, and each of its sub-parties referred to above, shall not engage in any form of sexual harassment ("SH"). SH is defined as any unwelcome conduct of a sexual nature that might reasonably be expected or be perceived to cause offense or humiliation, when such conduct interferes with work, is made a condition of employment or creates an intimidating, hostile or offensive work environment.

251. a) In the performance of the activities under this Project Document, the Implementing Partner shall (with respect to its own activities), and shall require from its sub-parties referred to in paragraph 4 (with respect to their activities) that they, have minimum standards and procedures in place, or a plan to develop and/or improve such standards and procedures in order to be able to take effective preventive and investigative action. These should include: policies on sexual harassment and sexual exploitation and abuse; policies on whistleblowing/protection against retaliation; and complaints, disciplinary and investigative mechanisms. In line with this, the Implementing Partner will and will require that such sub-parties will take all appropriate measures to:

    i. Prevent its employees, agents or any other persons engaged to perform any services under this Project Document, from engaging in SH or SEA;

    ii. Offer employees and associated personnel training on prevention and response to SH and SEA, where the Implementing Partner and its sub-parties referred to in paragraph 4 have not put in place its own training regarding the prevention of SH and SEA, the Implementing Partner and its sub-parties may use the training material available at UNDP;

    iii. Report and monitor allegations of SH and SEA of which the Implementing Partner and its sub-parties referred to in paragraph 4 have been informed or have otherwise become aware, and status thereof;

    iv. Refer victims/survivors of SH and SEA to safe and confidential victim assistance; and

    v. Promptly and confidentially record and investigate any allegations credible enough to warrant an investigation of SH or SEA. The Implementing Partner shall advise UNDP of any such allegations received and investigations being conducted by itself or any of its sub-parties referred to in paragraph 4 with respect to their activities under the Project Document, and shall keep UNDP informed during the investigation by it or any of such sub-parties, to the extent that such notification (i) does not jeopardize the conduct of the investigation, including but not limited to the safety or security of persons, and/or (ii) is not in contravention of any laws applicable to it. Following the investigation, the Implementing Partner shall advise UNDP of any actions taken by it or any of the other entities further to the investigation.

    b) The Implementing Partner shall establish that it has complied with the foregoing, to the satisfaction of UNDP, when requested by UNDP or any party acting on its behalf to provide such confirmation. Failure of the Implementing Partner, and each of its sub-parties referred to in paragraph 4, to comply of the foregoing, as determined by UNDP, shall be considered grounds for suspension or termination of the Project.

252. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

253. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any

concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

254. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

255. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, responsible parties, subcontractors and sub-recipients in implementing the project or using UNDP funds. The Implementing Partner will ensure that its financial management, anti-corruption and anti-fraud policies are in place and enforced for all funding received from or through UNDP.

256. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

257. In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP projects and programmes in accordance with UNDP's regulations, rules, policies and procedures. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.

258. The signatories to this Project Document will promptly inform one another in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

259. Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). The Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

260. UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail the Implementing Partner's obligations under this Project Document.

261. Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

262. *Note:* The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

263. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

264. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

265. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled "Risk Management Standard Clauses" are included, *mutatis mutandis*, in all sub-contracts or sub-agreements entered into further to this Project Document.

XII. ANNEXES